CHAPTER 06

Aldehydes and Ketones I

Content and rendering review packet

Original app content and actual Chromium screenshots. No textbook passages or textbook figures are embedded. Source labels and curriculum headings are not evidence of factual correctness.

Source snapshot SHA-256: c0859cc2751a1bed1e18fb12ee2b07b51aed79e9b259a394a7b2da771cb039cd

Review all claims, molecule graphs, lessons, exercises, hints and displayed states. Keys are author claims, not ground truth. Screenshots show static states with reduced motion: they do not prove animation timing or persistence behavior.

For every finding: item ID/page; severity (critical/major/minor); exact claim; derivation or reference evidence; proposed correction; confidence. List unreviewed/uncertain items. Model agreement is not scientific verification.

Coverage checklist

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ch06:lesson:carbonyl-redox:trap | carbonyl-redox:trap

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ch06:molecule:acetaldehyde | acetaldehyde

ch06:molecule:hemiacetal | hemiacetal

ch06:molecule:acetal | acetal

ch06:molecule:geminal | geminal

ch06:molecule:imine | imine

ch06:molecule:enamine | enamine

ch06:molecule:cyanohydrin | cyanohydrin

ch06:molecule:aceticacid | aceticacid

ch06:molecule:ethanol | ethanol

ch06:molecule:acetone | acetone

ch06:molecule:isopropanol | isopropanol

ch06:curriculum | scope-1, scope-2, scope-3

Curriculum scope

scope-1: 6.1 Description and Properties

scope-2: 6.2 Nucleophilic Addition Reactions

scope-3: 6.3 Oxidation–Reduction Reactions

Objective: Track carbonyl addition electron flow

Objective: Identify hydrates, hemiacetals, acetals, imines, and enamines

Objective: Predict carbonyl redox outcomes

Supplied Kaplan curriculum: Chapter 6; section scope checked against local body and summary | organic_chemistry_kaplan/OEBPS/chapter06b.xhtml

IUPAC Gold Book, acetals (search-index definition inspected; direct page fetch unavailable) | https://goldbook.iupac.org/terms/view/A00062

OpenStax 19.3, aldehyde oxidation | https://openstax.org/books/organic-chemistry/pages/19-3-oxidation-of-aldehydes-and-ketones

OpenStax 19.8, imines and enamines | https://openstax.org/books/organic-chemistry/pages/19-8-nucleophilic-addition-of-amines-imine-and-enamine-formation

No source passage is included. Request relevant authoritative excerpts when a claim cannot be checked by derivation.

carbonyl | Carbonyl polarity and geometry

Planar electrophile → tetrahedral addition product

carbonyl:rule | Core Rules

The carbonyl is polarized Cδ+–Oδ−. A nucleophile attacks carbon while the pi pair shifts to oxygen; protonation can then give an alcohol. Aldehydes are generally less crowded and less electron-rich than comparable ketones, so they often add nucleophiles more readily.

Visual card: Carbonyl Polarity, Bürgi-Dunitz Attack (~107°) & Aldehyde vs. Ketone Reactivity

Chemistry diagram: Carbonyl addition: encoded electron-flow arrows start at electrons and end at the accepting site.

C=O = sp^2 Planar (120°)

δ+ Carbon / δ− Oxygen

Aldehyde > Ketone Reactivity

Tetrahedral sp^3 Intermediate

Formaldehyde ($H_2C=O$) | Zero alkyl groups (least hindered) | Fastest nucleophilic addition

Aldehyde (R–CHO) | 1 alkyl group (less steric + less EDG) | More reactive than ketones

Ketone (R–CO–R′) | 2 alkyl groups (steric + 2 inductive EDGs) | Less electrophilic δ+ carbon

Permanent C=O Dipole & Resonance — — Oxygen is more electronegative than carbon ($R_2C=O \leftrightarrow R_2C^+–O^-$), making the sp^2 carbonyl carbon strongly electrophilic (δ+) and the oxygen weakly basic/nucleophilic (δ−).

Electronic Effect (Inductive Donation) — — Alkyl groups are electron-donating (EDG): a ketone has TWO alkyl groups donating electron density to reduce δ+ on carbon; an aldehyde has only ONE.

Steric Effect in the sp^3 Tetrahedral Intermediate — — Nucleophilic addition compresses bond angles from 120° (sp^2 trigonal planar) to 109.5° (sp^3 tetrahedral), crowding bulky ketone R groups.

carbonyl:cue | Memory Cues

Attack C; move pi electrons to O; then handle proton transfers. Aldehydes use -al, or -carbaldehyde when CHO is attached to a ring; ketones use -one. Use oxo- for a subordinate carbonyl. Methanal/formaldehyde, ethanal/acetaldehyde, propanal/propionaldehyde, butanal/butyraldehyde and pentanal/valeraldehyde are common name pairs; propanone is acetone.

Visual card: Acid-Catalyzed vs. Base-Catalyzed Carbonyl Addition Order

Are you running carbonyl nucleophilic addition in basic or acidic conditions?

Basic / Strong Nucleophile (RMgBr, $LiAlH_4$, CN^-, RO^-) — Step 1: Attack C=O → Step 2: Protonate Alkoxide (O^-) — Strong anionic nucleophiles attack the neutral C=O directly to form a tetrahedral alkoxide intermediate, then pick up H^+ on workup.

Acidic / Weak Nucleophile (ROH, H_2O + Catalytic H^+) — Step 1: Protonate C=O ($C=OH^+$) → Step 2: Attack! — Acid protonates the carbonyl oxygen FIRST, converting a moderate electrophile into a super-electrophilic oxocarbenium ion ($R_2C=OH^+$) so weak ROH can attack.

Comparing Carbonyl Boiling Points — Ketone/Aldehyde > Alkane/Ether, But < Alcohol — Strong C=O dipole-dipole interactions raise BP above alkanes/ethers, but lack of O–H means no self-H-bonding (lower BP than alcohols).

carbonyl:trap | Common Mistakes / MCAT Traps

Neutral aldehydes and ketones can accept hydrogen bonds but lack an O–H donor. Do not draw five bonds to carbon when a nucleophile adds. A planar carbonyl can be attacked from either face. At comparable mass and shape, boiling points often

follow alkane < aldehyde/ketone < alcohol; this is a trend, not a universal ordering across unrelated molecules.

Visual card: Drawing a Pentavalent Carbon or Forming O^- in Strong Acid

Common Trap — — Attacking C=O without breaking the π bond (5 bonds on C!), or drawing a strongly basic RO^- alkoxide intermediate in H_2SO_4.

Correct Reasoning — — Always push the C=O π electrons onto oxygen as the nucleophile attacks; keep intermediates cationic/neutral in acid and anionic/neutral in base!

carbonyl:app | MCAT Applications

A chiral environment can favor one face; without such bias, addition that creates a single new stereocenter often yields a racemic mixture.

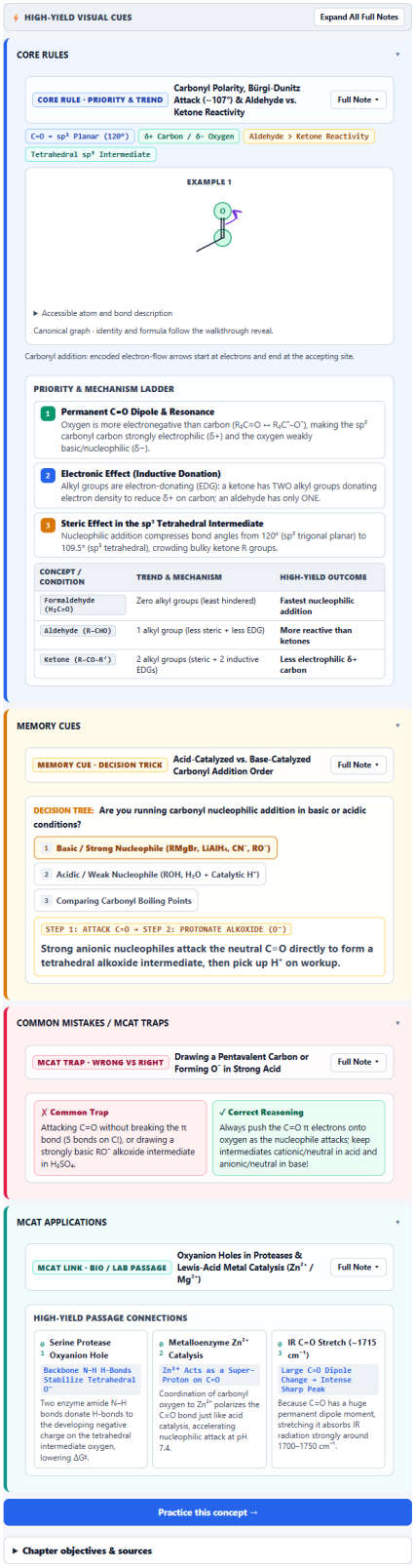
Visual card: Oxyanion Holes in Proteases & Lewis-Acid Metal Catalysis (Zn^{2+} / Mg^{2+})

Serine Protease Oxyanion Hole — Backbone N–H H-Bonds Stabilize Tetrahedral O^- — Two enzyme amide N–H bonds donate H-bonds to the developing negative charge on the tetrahedral intermediate oxygen, lowering $\Delta G^{\ddagger}$.

Metalloenzyme Zn^{2+} Catalysis — Zn^{2+} Acts as a Super-Proton on C=O — Coordination of carbonyl oxygen to Zn^{2+} polarizes the C=O bond just like acid catalysis, accelerating nucleophilic attack at pH 7.4.

IR C=O Stretch (~1715 cm^{-1}) — Large C=O Dipole Change → Intense Sharp Peak — Because C=O has a huge permanent dipole moment, stretching it absorbs IR radiation strongly around 1700–1750 cm^{-1}.

Screenshot: carbonyl-visuals



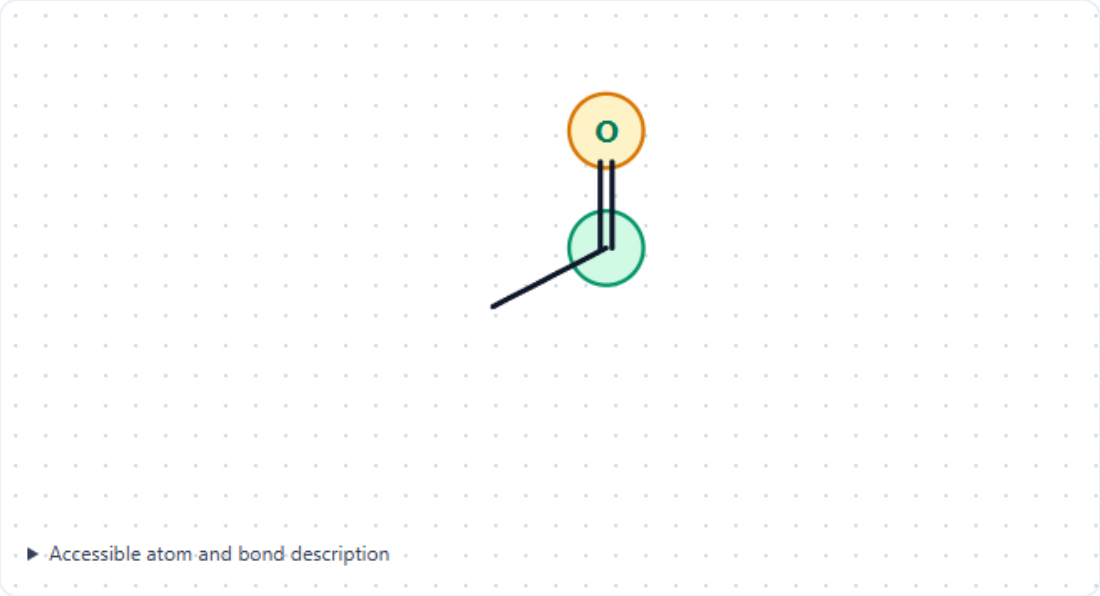
carbonyl:step-1 | 1. Find the electrophile

The C=O carbon is trigonal planar and electron poor. Oxygen is electron rich.

Displayed molecule ID: acetaldehyde

{"highlights": {"prin": [1], "sub": [2]}, "revealName": false}

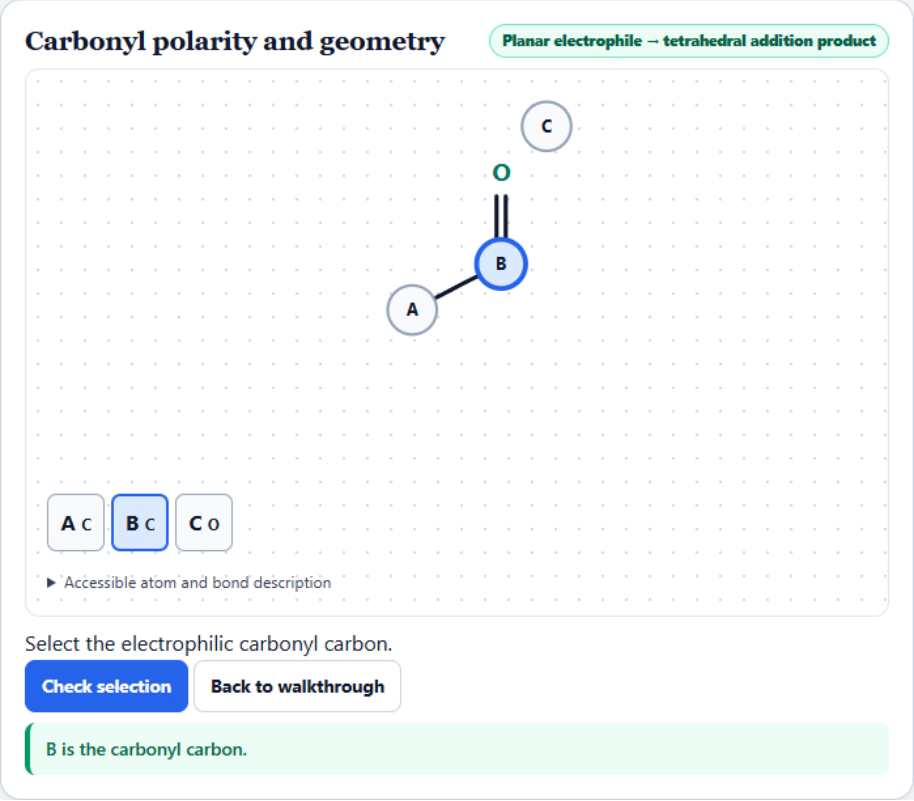
Screenshot: carbonyl-step-1



Embedded challenge: Select the electrophilic carbonyl carbon.

{"prompt": "Select the electrophilic carbonyl carbon.", "answer": [1], "explanation": "B is the carbonyl carbon."}

Screenshot: carbonyl-step-1-challenge



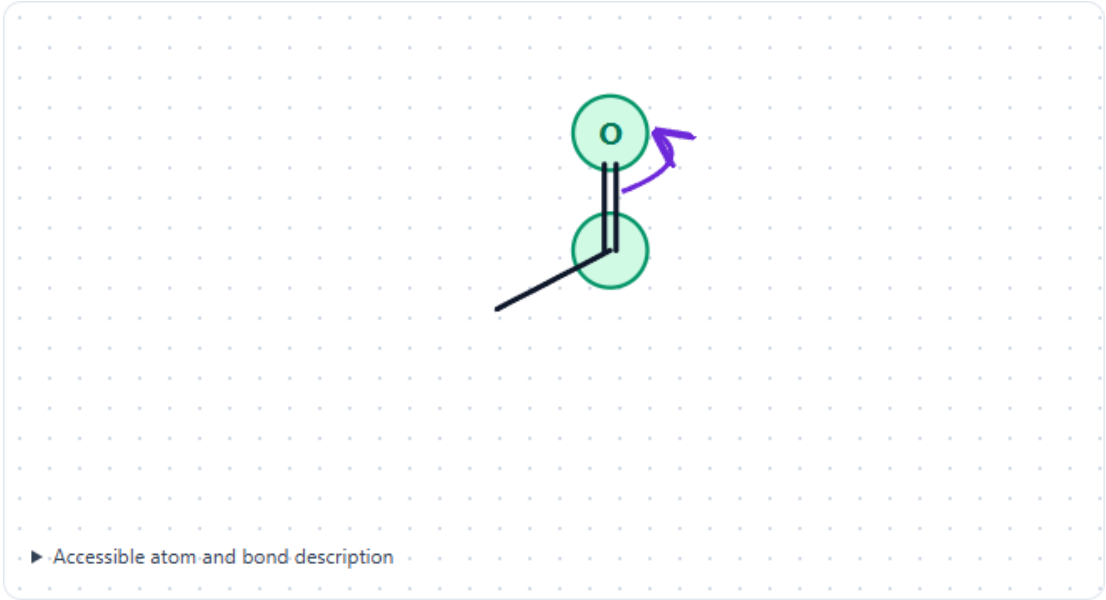
carbonyl:step-2 | 2. Move the pi pair

The purple curved arrow shows the carbonyl pi pair moving toward oxygen during nucleophilic attack. This arrow is one part of an addition mechanism; the incoming nucleophile is described in the lesson rather than drawn here.

Displayed molecule ID: acetaldehyde

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Screenshot: carbonyl-step-2



oxygen-addition | Hydrates, hemiacetals, and acetals

Distinguish OH from OR on the same carbon

oxygen-addition:rule | Core Rules

A hydrate has two OH groups on one carbon. In traditional terminology, aldehyde-derived hemiacetals have OH, OR, H and R at the former carbonyl carbon; further acid-catalyzed reaction gives acetals with two OR groups, H and R. Ketone-derived counterparts have two carbon groups and are called hemiketals (OH + OR) and ketals (two OR). Broad IUPAC acetal/hemiacetal classes also include the ketone-derived compounds. Equilibria depend on water/alcohol activities.

Visual card: Hydrates, Hemiacetals/Hemiketals & Acid-Only Acetal Protecting Groups

Chemistry diagram: Compare hemiacetal, ketal and hydrate connectivity. These are separate illustrative examples.

1 Eq ROH → Hemiacetal (RO–C–OH)

2 Eq ROH + H^+ (-H_2O) → Acetal (RO–C–OR)

Acetals = Base-Stable Protecting Groups!

Hydrate (Gem-diol) | $R_2C(OH)_2$ (Two –OH on same C) | Reversible in H^+ OR HO^-; favors C=O

Hemiacetal / Hemiketal | $R_2C(OH)(OR')$ (1 –OH + 1 –OR') | Forms in acid OR base; cyclic sugars stable!

Acetal / Ketal | $R_2C(OR')_2$ (Two –OR' on same C) | Forms ONLY in ACID; locked in base/RMgBr/$LiAlH_4$!

Hemiacetal / Hemiketal Formation (1 Eq ROH) — — Acid- or base-catalyzed addition of 1 ROH across C=O yields a carbon bonded to ONE –OH and ONE –OR (unstable open-chain, but favored in 5/6-membered sugar rings!).

Acetal / Ketal Formation Requires ACID (2 Eq ROH + H^+, −H_2O) — — Protonation turns the hemiacetal –OH into –OH_2^+, which leaves to form a resonance-stabilized oxocarbenium ion ($R_2C{=}OR'^+$) attacked by a second ROH.

Base Immunity & Acid Deprotection — — Because RO^- cannot leave in base, acetals are 100% stable to NaOH, $LiAlH_4$, $NaBH_4$, and Grignards; hydrolyze back to C=O + 2 ROH with aqueous acid (H_3O^+).

oxygen-addition:cue | Memory Cues

Two OH = hydrate; OH + OR = hemiacetal; two OR = acetal. Ketone-derived examples are also called hemiketals/ketals.

Visual card: Identify the Carbonyl Derivative by Counting Oxygen Attachments on the Same Carbon

Inspect the sp^3 carbon that holds two single bonds to oxygen:

Bonded to –OH and –OR (plus R/H) — Hemiacetal (or Hemiketal if from Ketone) — Examples: α/β-D-glucopyranose at anomeric C1! Equilibrates with the open-chain aldehyde in water (mutarotation, reducing sugar).

Bonded to –OR and –OR (Two Ether Links on 1 C) — Acetal (or Ketal) — Base-Stable Lock! — Examples: sucrose/starch glycosidic bonds or ethylene glycol dioxolane protecting group; NON-reducing and stable to base/$LiAlH_4$!

Bonded to ONLY ONE –OR (C–O–C on Separate Carbons) — Simple Ether (NOT an Acetal) — Ordinary ethers do not hydrolyze in mild aqueous acid, whereas acetals rapidly revert to aldehydes/ketones in H_3O^+.

oxygen-addition:trap | Common Mistakes / MCAT Traps

Acetal formation requires acid catalysis; base alone does not turn OH− into a good leaving group. Cyclic sugar hemiacetals can be stable because intramolecular ring formation is favorable.

Visual card: Trying to Form or Hydrolyze an Acetal in Base (NaOH / RO^-)

Common Trap — — Writing RCHO + 2 CH_3OH + NaOH → Acetal, or cleaving a glycosidic acetal bond with aqueous NaOH.

Correct Reasoning — — Base catalysis STOPS at the hemiacetal (HO^- cannot leave to form an oxocarbenium ion)! Acetals form and hydrolyze ONLY in acid (H^+).

oxygen-addition:app | MCAT Applications

Monosaccharides cyclize through hemiacetal or hemiketal formation. The new anomeric center can have two configurations. Acetal/ketal protection can preserve a carbonyl during many basic or hydride reactions, including suitable LiAlH4 conditions. Aqueous acid restores the carbonyl; compatibility must be checked for the whole substrate.

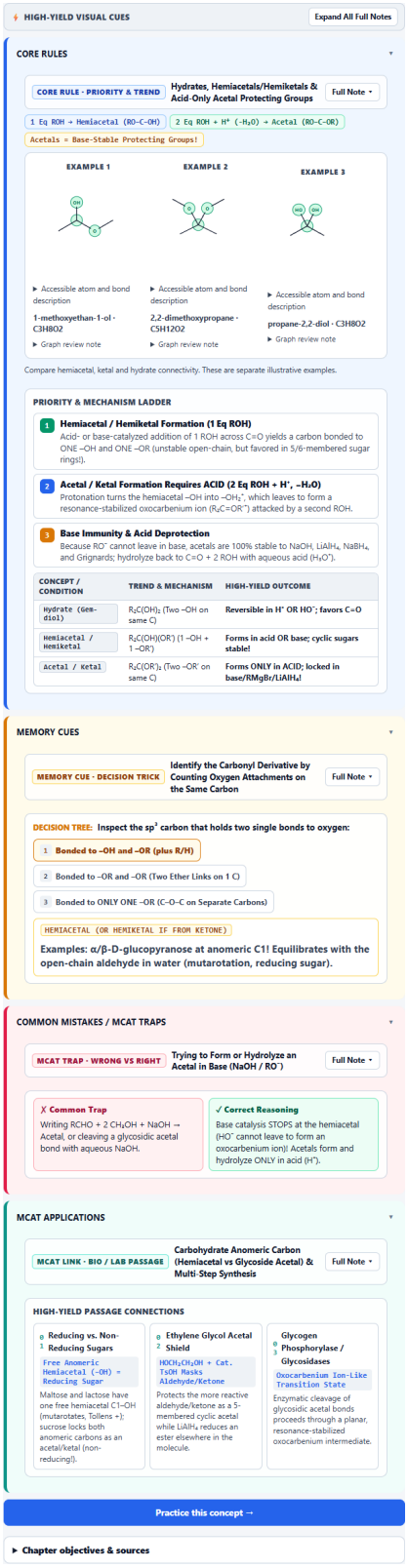
Visual card: Carbohydrate Anomeric Carbon (Hemiacetal vs Glycoside Acetal) & Multi-Step Synthesis

Reducing vs. Non-Reducing Sugars — Free Anomeric Hemiacetal (–OH) = Reducing Sugar — Maltose and lactose have one free hemiacetal C1–OH (mutarotates, Tollens +); sucrose locks both anomeric carbons as an acetal/ketal (non-reducing!).

Ethylene Glycol Acetal Shield — $HOCH_2CH_2OH$ + Cat. TsOH Masks Aldehyde/Ketone — Protects the more reactive aldehyde/ketone as a 5-membered cyclic acetal while $LiAlH_4$ reduces an ester elsewhere in the molecule.

Glycogen Phosphorylase / Glycosidases — Oxocarbenium Ion-Like Transition State — Enzymatic cleavage of glycosidic acetal bonds proceeds through a planar, resonance-stabilized oxocarbenium intermediate.

Screenshot: oxygen-addition-visuals



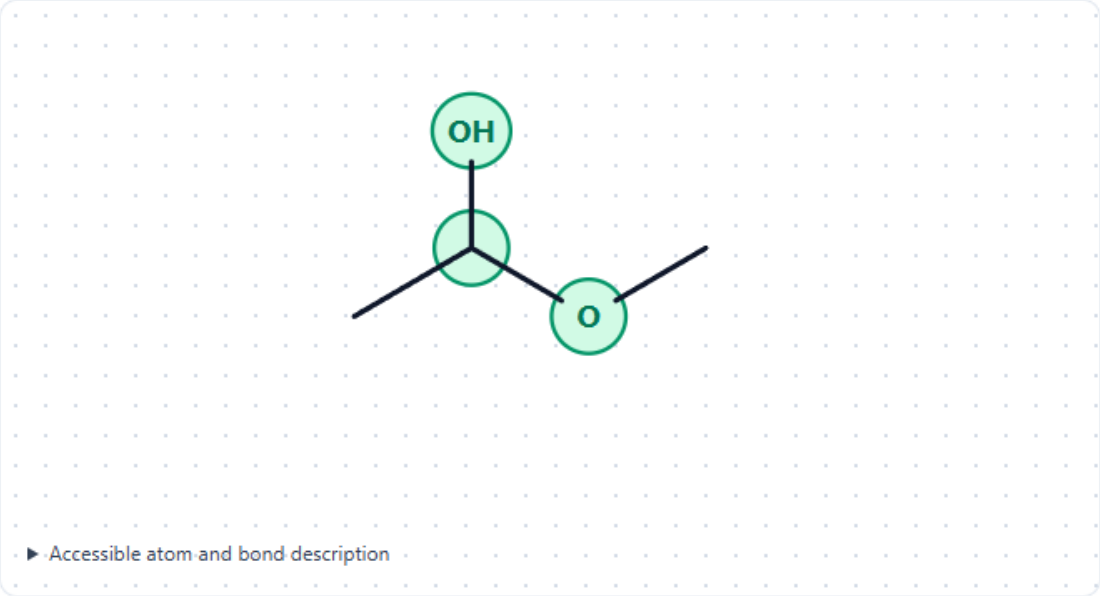
oxygen-addition:step-1 | 1. Identify a hemiacetal

This carbon bears both OH and OCH3, along with methyl and H. It is a hemiacetal from ethanal and methanol.

Displayed molecule ID: hemiacetal

{"highlights": {"prin": [1, 2, 3]}, "revealName": true}

Screenshot: oxygen-addition-step-1



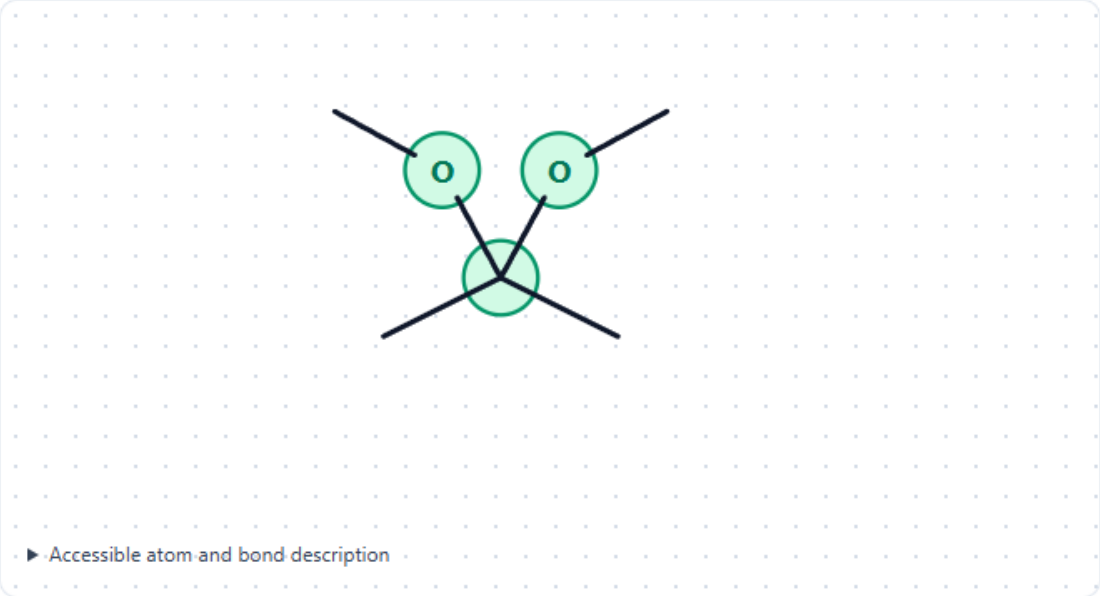
oxygen-addition:step-2 | 2. Compare a ketal

This separate ketone-derived example is a ketal: two OCH3 groups and two carbon groups surround the central carbon. It is not the direct next product of the ethanal hemiacetal in step 1. Aqueous-acid hydrolysis regenerates a ketone.

Displayed molecule ID: acetal

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Screenshot: oxygen-addition-step-2



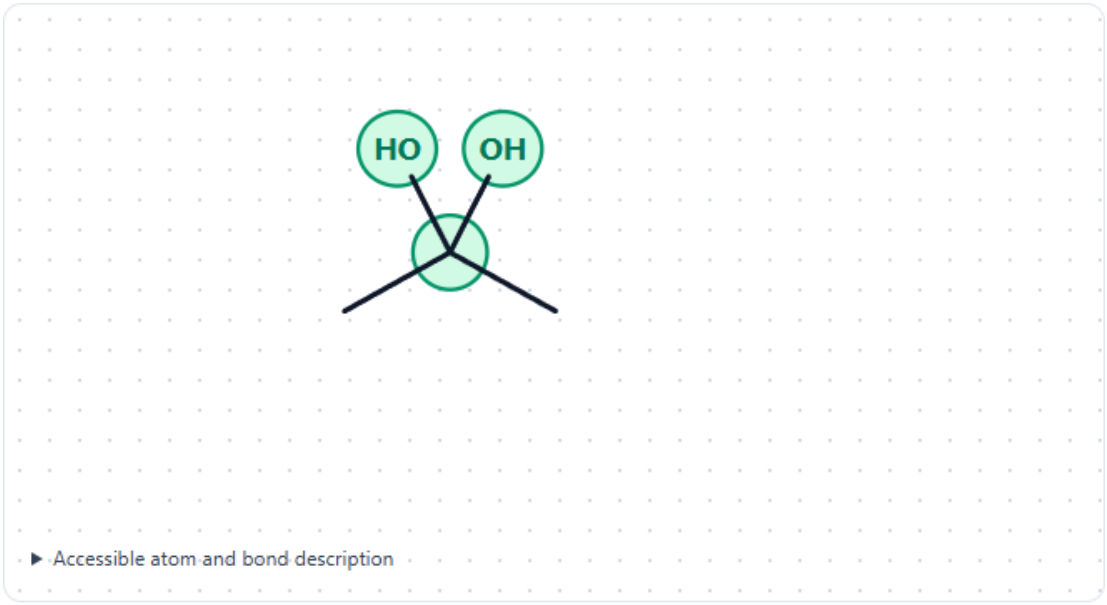
oxygen-addition:step-3 | 3. Compare a hydrate

Here both oxygen substituents are OH: a geminal diol. The equilibrium does not equally favor hydrates for every carbonyl.

Displayed molecule ID: geminal

{"highlights": {"prin": [1, 3, 4]}, "revealName": true}

Screenshot: oxygen-addition-step-3



nitrogen-addition | Imines, enamines, and cyanohydrins

Reagent identity changes the product

nitrogen-addition:rule | Core Rules

A primary amine condenses with an aldehyde or ketone to form an imine after addition, proton transfers, and water loss. A secondary amine can form an enamine when an alpha H is available. Related nitrogen reagents form oximes (hydroxylamine), hydrazones (hydrazine derivatives), and semicarbazones (semicarbazide). Cyanide addition followed by protonation forms a cyanohydrin and creates a C–C bond. Ammonia can also give an imine under suitable condensation conditions. An imine with an available alpha H can interconvert with an enamine tautomer by proton and double-bond relocation. Secondary-amine condensation proceeds through an iminium ion and alpha deprotonation; no N–H remains at that intermediate.

Visual card: Imine and enamine formation: amine class, acid catalysis and competing protonation

Chemistry diagram: Separate imine and enamine examples: different skeletons, not a displayed tautomer pair.

1° Amine (RNH_2) → Imine (C=NR)

2° Amine (R_2NH) → Enamine (C=C–NR_2)

Mild Acid optimum pH depends on substrate

HCN/CN^- → Cyanohydrin

1° Amine (RNH_2) | Has 2 N–H protons | Loses H_2O → Imine (Schiff Base, $R_2C{=}NR'$)

2° Amine (R_2NH) | Has only 1 N–H proton | Deprotonates α-C → Enamine ($RCH{=}CR{-}NR'_2$)

Cyanide (CN^- / HCN) | Reversible C–C bond formation | Forms α-hydroxy nitrile (Cyanohydrin)

Primary Amine (RNH_2 / NH_2OH / NH_2NH_2) → Imine (C=N) — — Nucleophilic addition forms a carbinolamine; acid protonates –OH to $–OH_2^+$, and the nitrogen lone pair pushes out water to form $R_2C{=}NR'$ (imine/oxime/hydrazone).

Secondary Amine (R_2NH) → Enamine (C=C–NR_2) — — After losing H_2O, the iminium ion ($R_2C{=}NR'_2{}^+$) has NO proton left on nitrogen! Base removes an α-C–H proton instead to form a nucleophilic alkene-amine (enamine).

Cyanohydrin Formation (KCN / HCN) — — CN^- attacks the carbonyl to form a new C–C bond ($R_2C(OH)CN$); nitrile can then be reduced to a 1° amine or hydrolyzed to an α-hydroxy carboxylic acid!

nitrogen-addition:cue | Memory Cues

Primary amine → C=N; secondary amine + alpha H → C=C–N. Cyanide adds its carbon.

Visual card: Balance nucleophile availability with acid-catalyzed dehydration

What happens if the pH is too high or too low during imine formation?

Too Basic (pH > 9): No H^+ Available — Carbinolamine –OH Cannot Leave as H_2O! — Amine attacks C=O, but without acid to protonate –OH into $–OH_2^+$, dehydration stalls completely.

Too Acidic (pH < 2): All RNH_2 Protonated to RNH_3^+ — No Nucleophilic Lone Pair Left to Attack! — Strong acid converts 100% of the amine into non-nucleophilic ammonium ions (RNH_3^+), stopping Step 1.

Sweet Spot (substrate-dependent pH) — Enough Free RNH_2 to Attack + Enough H^+ to Dehydrate — Produces the classic bell-shaped rate vs. pH profile tested on the MCAT!

nitrogen-addition:trap | Common Mistakes / MCAT Traps

Too much acid protonates the amine and suppresses its nucleophilicity. Enamine formation needs an alpha hydrogen. Do not attach cyanide through nitrogen in a standard cyanohydrin product.

Visual card: Drawing a Charged Iminium or Enamine from the Wrong Amine Class

Common Trap — — Forming a neutral $C=NR_2$ imine from a secondary amine, or calling separate imine and enamine molecules from different amines "tautomers".

Correct Reasoning — — 1° amines have 2 N–H protons → neutral Imine (C=NR); 2° amines have 1 N–H proton → neutral Enamine ($C=C–NR_2$). (Only an imine and enamine with the SAME formula interconvert as tautomers).

nitrogen-addition:app | MCAT Applications

Schiff-base (imine) intermediates help enzymes reorganize electron density; nitrogen chemistry is a bridge to amino-acid metabolism.

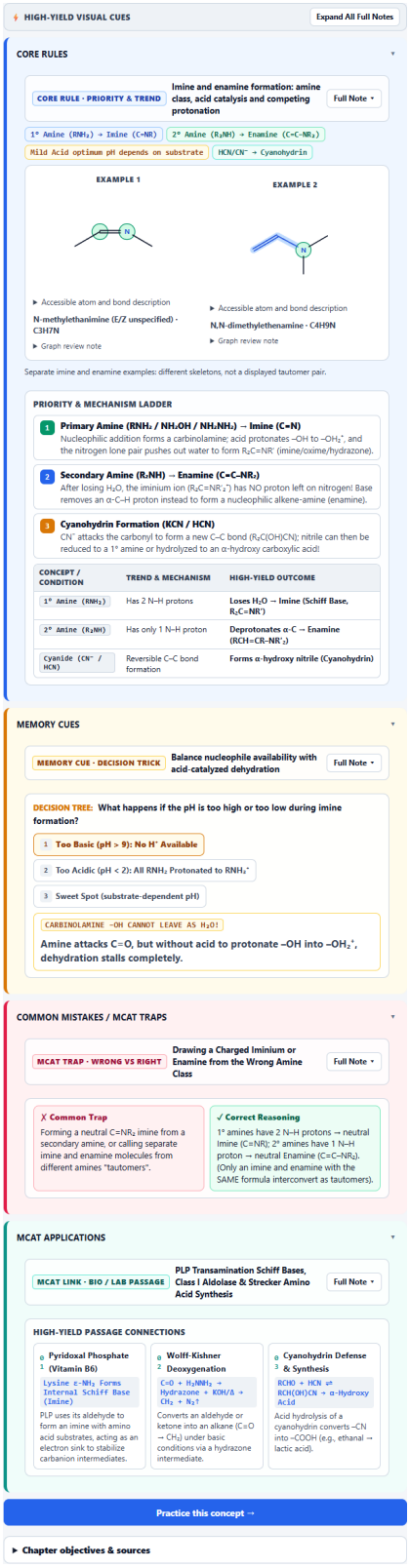
Visual card: PLP Transamination Schiff Bases, Class I Aldolase & Strecker Amino Acid Synthesis

Pyridoxal Phosphate (Vitamin B6) — Lysine ε-NH_2 Forms Internal Schiff Base (Imine) — PLP uses its aldehyde to form an imine with amino acid substrates, acting as an electron sink to stabilize carbanion intermediates.

Wolff-Kishner Deoxygenation — C=O + H_2NNH_2 → Hydrazone + KOH/Δ → CH_2 + N_2↑ — Converts an aldehyde or ketone into an alkane (C=O → CH_2) under basic conditions via a hydrazone intermediate.

Cyanohydrin Defense & Synthesis — RCHO + HCN ⇌ RCH(OH)CN → α-Hydroxy Acid — Acid hydrolysis of a cyanohydrin converts –CN into –COOH (e.g., ethanal → lactic acid).

Screenshot: nitrogen-addition-visuals



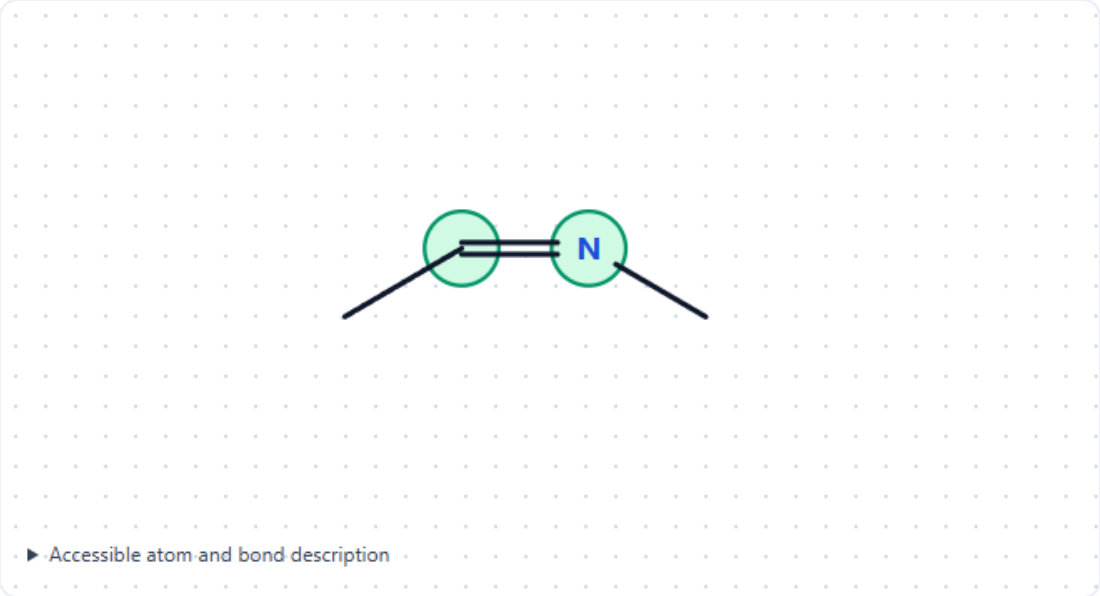
nitrogen-addition:step-1 | 1. Imine bond

The C=N bond replaces the starting C=O after condensation with a primary amine. This connectivity drawing does not assign E/Z about C=N.

Displayed molecule ID: imine

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Screenshot: nitrogen-addition-step-1



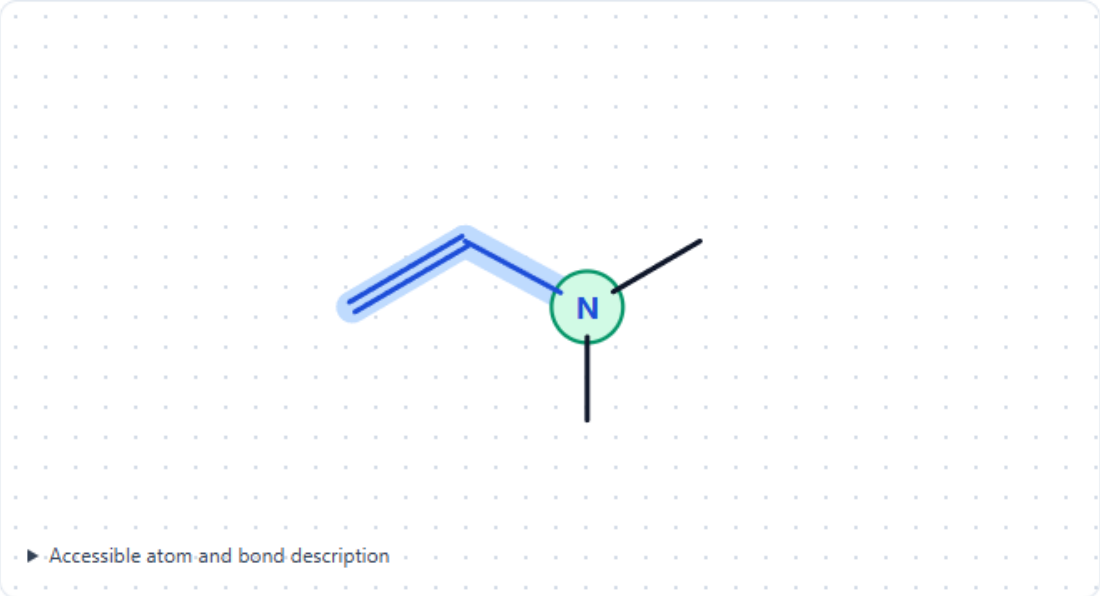
nitrogen-addition:step-2 | 2. Enamine bond

An enamine has N attached to a C=C system. The nitrogen lone pair is conjugated with the double bond, giving nucleophilic carbon character. The displayed imine and enamine are separate examples with different substituents and formulas, not a tautomeric pair.

Displayed molecule ID: enamine

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Screenshot: nitrogen-addition-step-2



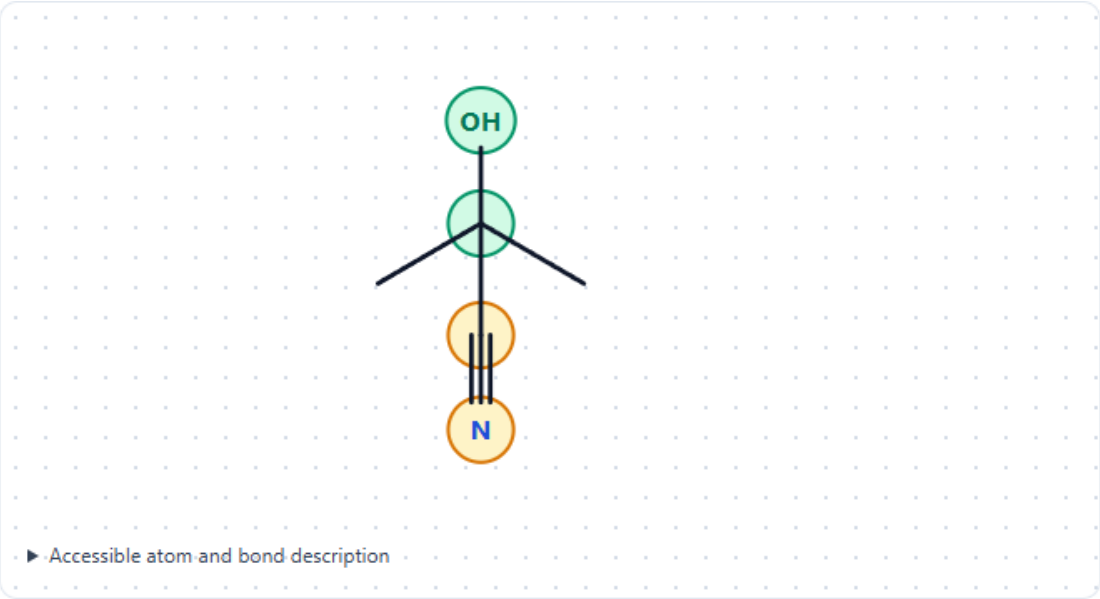
nitrogen-addition:step-3 | 3. Cyanohydrin carbon count

Acetone cyanohydrin contains a new C–C bond between the former carbonyl carbon and the nitrile carbon. The nitrile N is triple-bonded to that new carbon.

Displayed molecule ID: cyanohydrin

{"highlights": {"prin": [1, 3], "sub": [4, 5]}, "revealName": true}

Screenshot: nitrogen-addition-step-3



carbonyl-redox | Aldehyde oxidation and hydride reduction

Separate addition from net redox

carbonyl-redox:rule | Core Rules

Aldehydes can be oxidized to carboxylic acids; ordinary ketones resist analogous oxidation without C–C cleavage. NaBH4 and LiAlH4 reduce aldehydes to primary alcohols and ketones to secondary alcohols. Hydride adds first, then oxygen is protonated in workup or by the medium. Suitable aqueous KMnO4 or Cr(VI) reagents (CrO3/acid or dichromate/acid) oxidize aldehydes. Tollens reagent is ammoniacal Ag(I), which is reduced to silver while an aldehyde becomes carboxylate in the basic medium; acid workup gives the acid. H2O2 can oxidize aldehydes under suitable conditions. Anhydrous PCC is commonly chosen to make aldehydes from primary alcohols while limiting overoxidation; it is not the only selective reagent.

Visual card: Diagnostic Aldehyde Oxidation (Tollens/Benedict) & Hydride Reduction ($NaBH_4$ vs. $LiAlH_4$)

Chemistry diagram: Aldehyde reduction to primary alcohol with a suitable hydride reagent, then protonation; schematic, not balanced.

Tollens $[Ag(NH_3)_2]^+$ → Silver Mirror (Aldehyde Only!)

$NaBH_4$ = Mild (Ald/Ket Only)

$LiAlH_4$ = Fierce (All C=O)

Tollens / Benedict Test | Oxidizes Aldehydes (RCHO → $RCOO^-$) | Ketones give NEGATIVE test (unless α-hydroxy tautomerizes)

$NaBH_4$ (in MeOH/H_2O) | Reduces Aldehydes (→ 1° ROH) & Ketones (→ 2° ROH) | Leaves Esters, Carboxylic Acids & Amides UNTOUCHED

$LiAlH_4$ (Anhydrous Et_2O) | Reduces Ald, Ket, Esters, Acids & Amides | Violently reacts with water; requires 2nd step H_3O^+ workup

Aldehyde-Specific Oxidation (Tollens, Fehling, Benedict) — — Aldehydes have a weak C–H bond on the carbonyl carbon and oxidize easily to carboxylates ($Ag^+ \rightarrow Ag^0$ mirror; Cu^{2+} blue → Cu_2O brick-red precipitate); simple ketones do NOT react.

Chemoselective Hydride Reduction ($NaBH_4$ vs. $LiAlH_4$) — — Al is less electronegative than B, making the Al–H bond far more polar and hydride-donating: $LiAlH_4$ reduces esters/acids/amides, while mild $NaBH_4$ reduces only aldehydes/ketones (and acid chlorides).

Stereochemistry of Achiral Ketone Reduction — — Hydride attacks the planar sp^2 ketone equally from the top (Re) and bottom (Si) faces, yielding a 50:50 RACEMIC mixture if a new stereocenter forms.

carbonyl-redox:cue | Memory Cues

Aldehyde reduction → 1° alcohol; ketone reduction → 2° alcohol.

Visual card: Carbonyl Redox Reagent & Stereochemical Outcome Selector

What transformation or diagnostic test is being performed?

Distinguish Aldehyde from Ketone in Lab — Add Tollens Reagent ($[Ag(NH_3)_2]^+$) → Silver Mirror — Aldehyde is oxidized to $RCOO^-$ while Ag^+ is reduced to metallic Ag(s). (Note: α-hydroxy ketones in sugars give a positive test via enediol tautomerization!).

Reduce Ketone to 2° ROH Without Touching an Ester — Use $NaBH_4$ in Ethanol — Mild $NaBH_4$ selectively reduces the ketone C=O while the ester –COOR remains intact.

Track Where H Comes From in $LiAlD_4$ then H_2O — Deuterium (D) Bonds to Carbon; H Bonds to Oxygen! — Hydride/deuteride ($D:^-$) attacks the electrophilic carbonyl CARBON; the aqueous workup protonates the alkoxide OXYGEN.

carbonyl-redox:trap | Common Mistakes / MCAT Traps

Do not predict an alkane from ordinary hydride reduction of C=O. The oxygen remains in the alcohol. A positive aldehyde oxidation test is contextual evidence, not a complete structural identification. Tollens reagent is not a general primary-alcohol oxidant. Water, pH, workup and other oxidizable groups affect selectivity.

Visual card: Mixing $LiAlH_4$ with Water/Alcohol in Step 1 or Putting Workup H^+ on Carbon

Common Trap — — Writing "$LiAlH_4$ in MeOH" in a single step, or putting the H from H_3O^+ onto the carbonyl carbon.

Correct Reasoning — — $LiAlH_4$ reacts explosively with protic solvents ($H:^- + H_2O \rightarrow H_2\uparrow$); always run Step 1 in anhydrous ether/THF and Step 2 H_3O^+ workup (which protonates O^-).

carbonyl-redox:app | MCAT Applications

Biological NADH-dependent carbonyl reductions transfer a hydride equivalent; the enzyme can control stereochemistry.

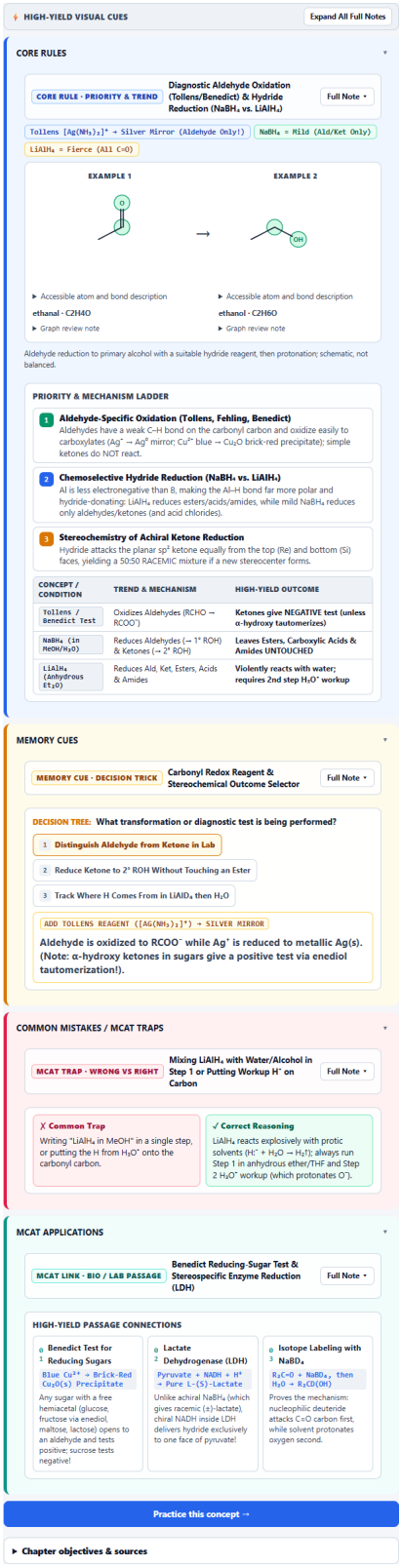
Visual card: Benedict Reducing-Sugar Test & Stereospecific Enzyme Reduction (LDH)

Benedict Test for Reducing Sugars — Blue Cu^{2+} → Brick-Red $Cu_2O(s)$ Precipitate — Any sugar with a free hemiacetal (glucose, fructose via enediol, maltose, lactose) opens to an aldehyde and tests positive; sucrose tests negative!

Lactate Dehydrogenase (LDH) — Pyruvate + NADH + H^+ → Pure L-(S)-Lactate — Unlike achiral $NaBH_4$ (which gives racemic (±)-lactate), chiral NADH inside LDH delivers hydride exclusively to one face of pyruvate!

Isotope Labeling with $NaBD_4$ — $R_2C=O$ + $NaBD_4$, then H_2O → $R_2CD(OH)$ — Proves the mechanism: nucleophilic deuteride attacks C=O carbon first, while solvent protonates oxygen second.

Screenshot: carbonyl-redox-visuals



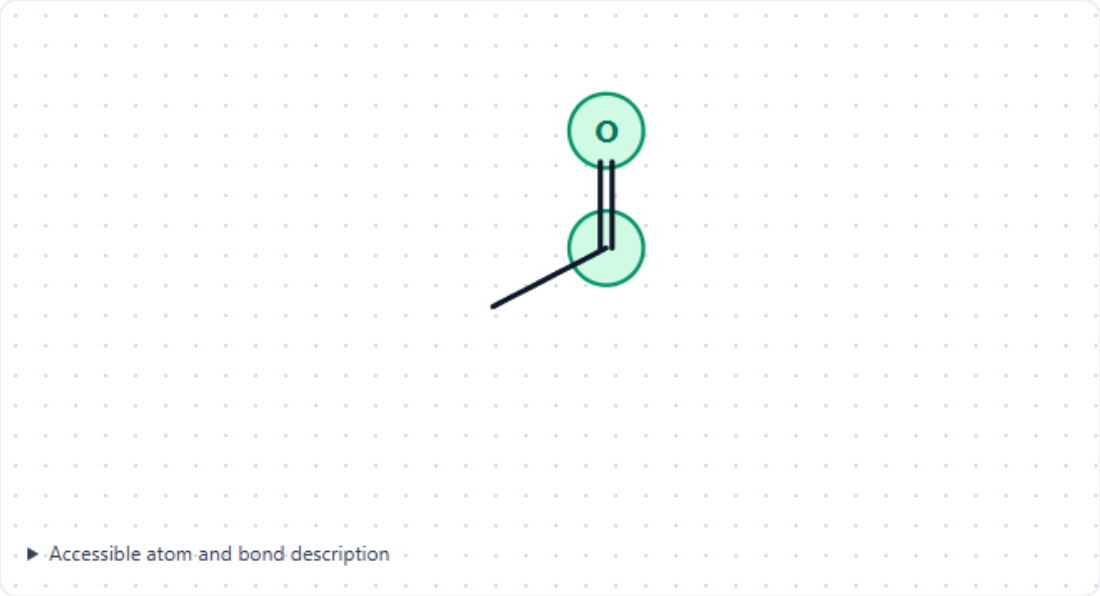
carbonyl-redox:step-1 | 1. Aldehyde substrate

Ethanal has a hydrogen on its carbonyl carbon, allowing ordinary oxidation to a carboxylic acid.

Displayed molecule ID: acetaldehyde

{"highlights": {"prin": [1, 2]}, "revealName": true}

Screenshot: carbonyl-redox-step-1



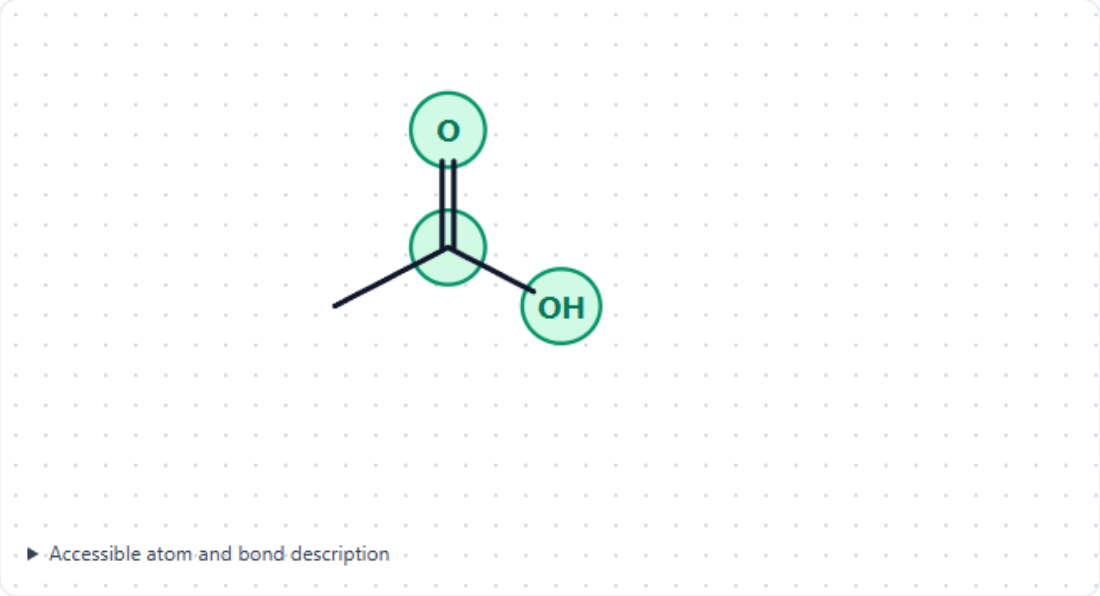
carbonyl-redox:step-2 | 2. Oxidized product

Ethanoic acid retains the two-carbon skeleton and adds a carbon–oxygen bond overall.

Displayed molecule ID: aceticacid

{"highlights": {"prin": [1, 2, 3]}, "revealName": true}

Screenshot: carbonyl-redox-step-2



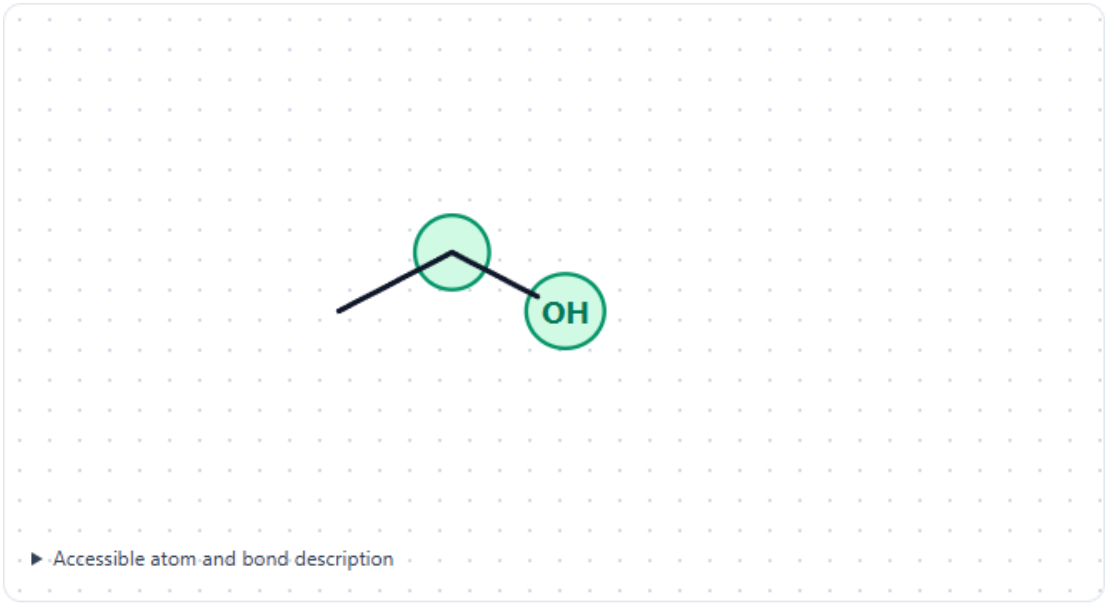
carbonyl-redox:step-3 | 3. Reduced product

The reduction product ethanol retains the same oxygen but has a single C–O bond and more C–H bonding.

Displayed molecule ID: ethanol

{"highlights": {"prin": [1, 2]}, "revealName": true}

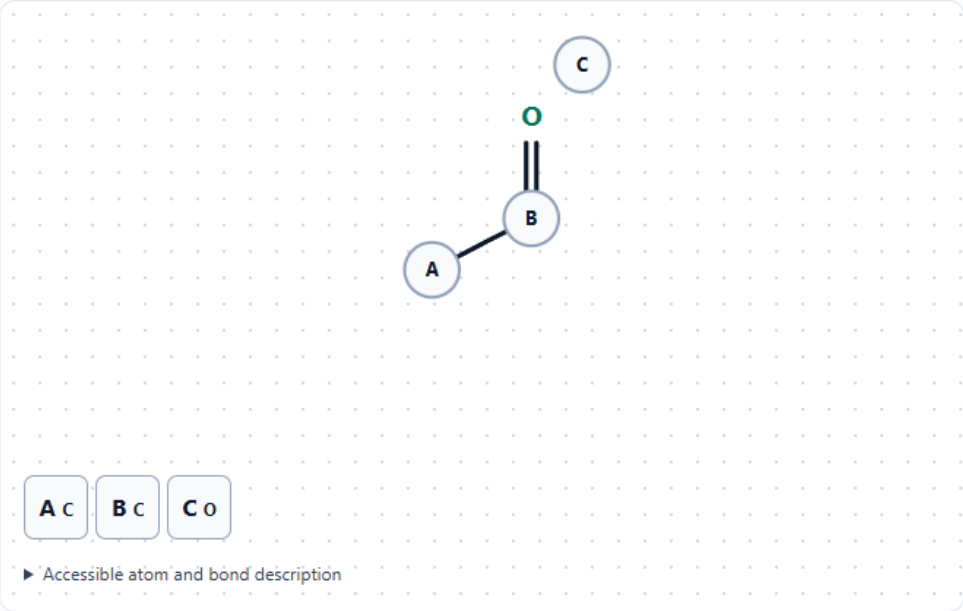
Screenshot: carbonyl-redox-step-3



addition-site | Follow nucleophilic addition

The canvas retains the starting ethanal to trace one electron-flow step. The incoming nucleophile and tetrahedral product are not drawn.

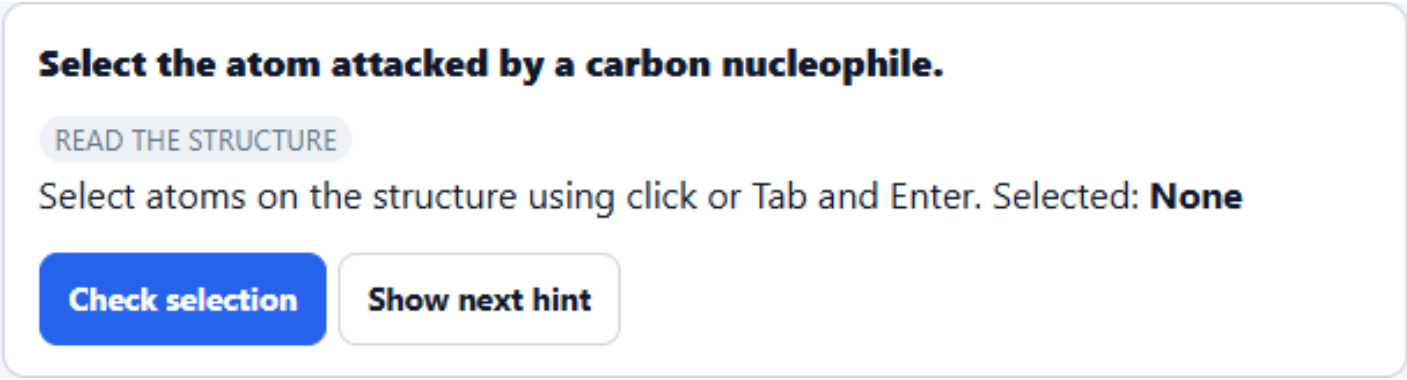
Screenshot: addition-site-stage-1-molecule



addition-site:stage-1 | Select the atom attacked by a carbon nucleophile.

Answer type: atoms

Screenshot: addition-site-stage-1-question



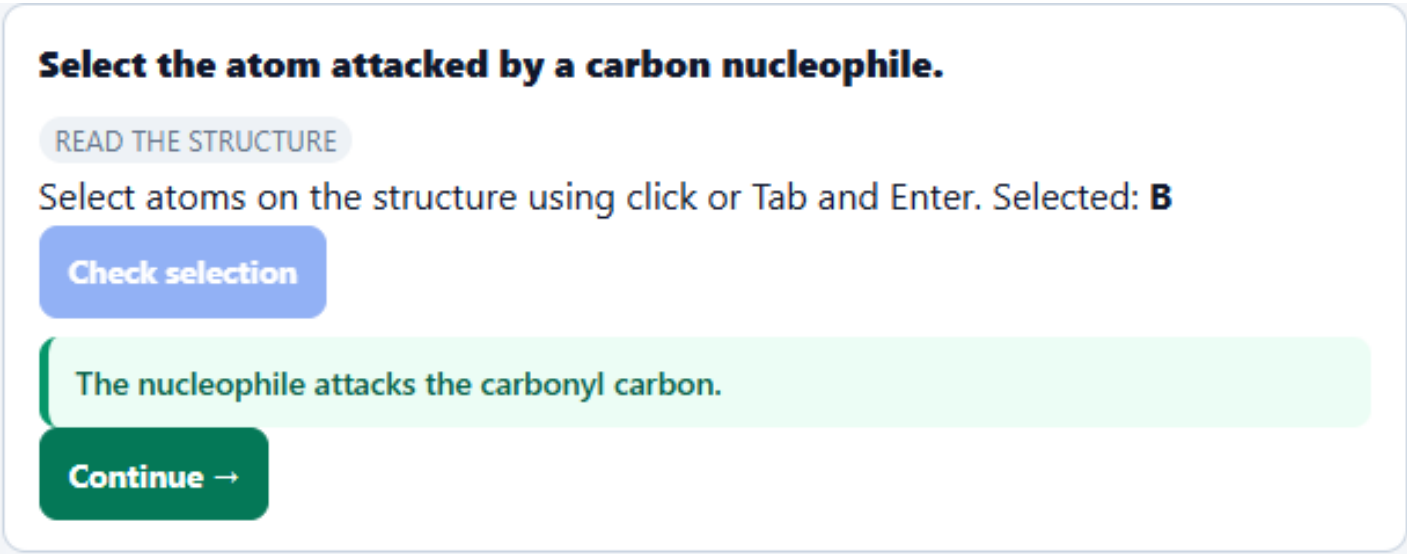
{"authorAnswer": [1], "tolerance": null, "highlightsAfterCorrectAnswer": {"prin": [1]}}

Hint 1: Locate the positively polarized end of C=O.

Hint 2: Carbon is less electronegative than oxygen.

Explanation: The nucleophile attacks the carbonyl carbon.

Screenshot: addition-site-stage-1-feedback



addition-site:stage-2 | As the new sigma bond forms, the C=O pi pair moves to...

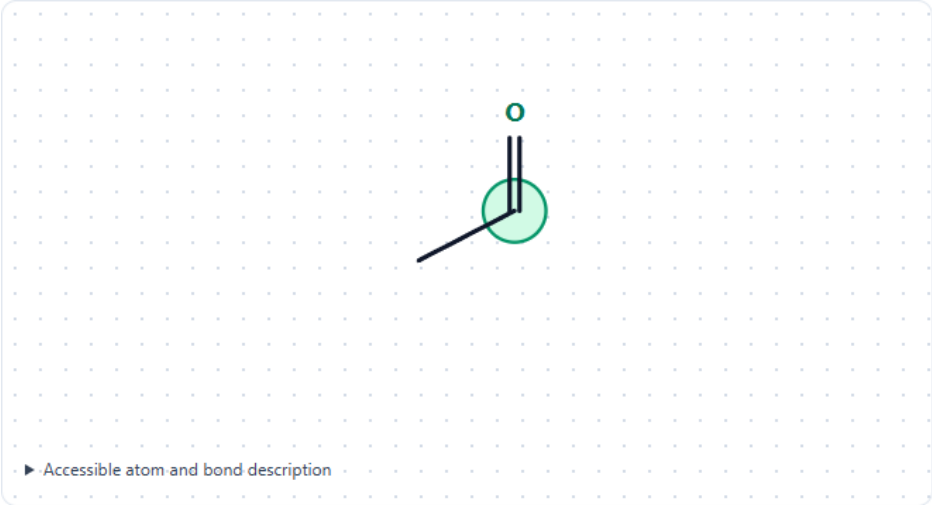
Answer type: choice

Option 0 (A): Oxygen

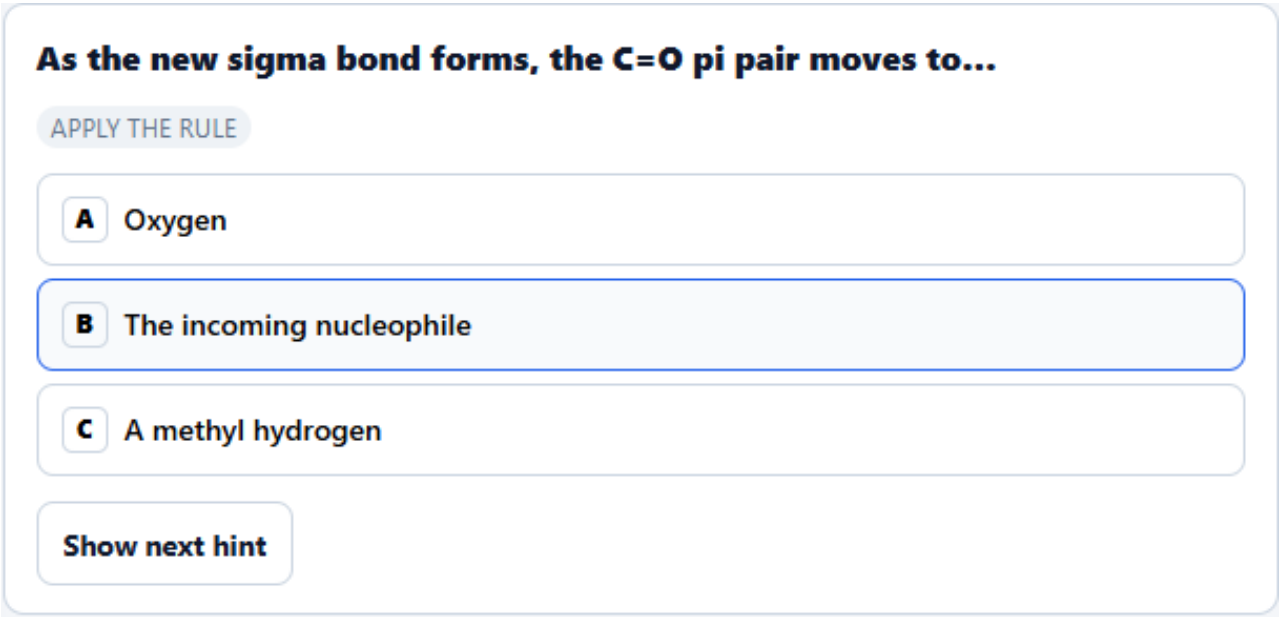
Option 1 (B): The incoming nucleophile

Option 2 (C): A methyl hydrogen

Screenshot: addition-site-stage-2-molecule



Screenshot: addition-site-stage-2-question



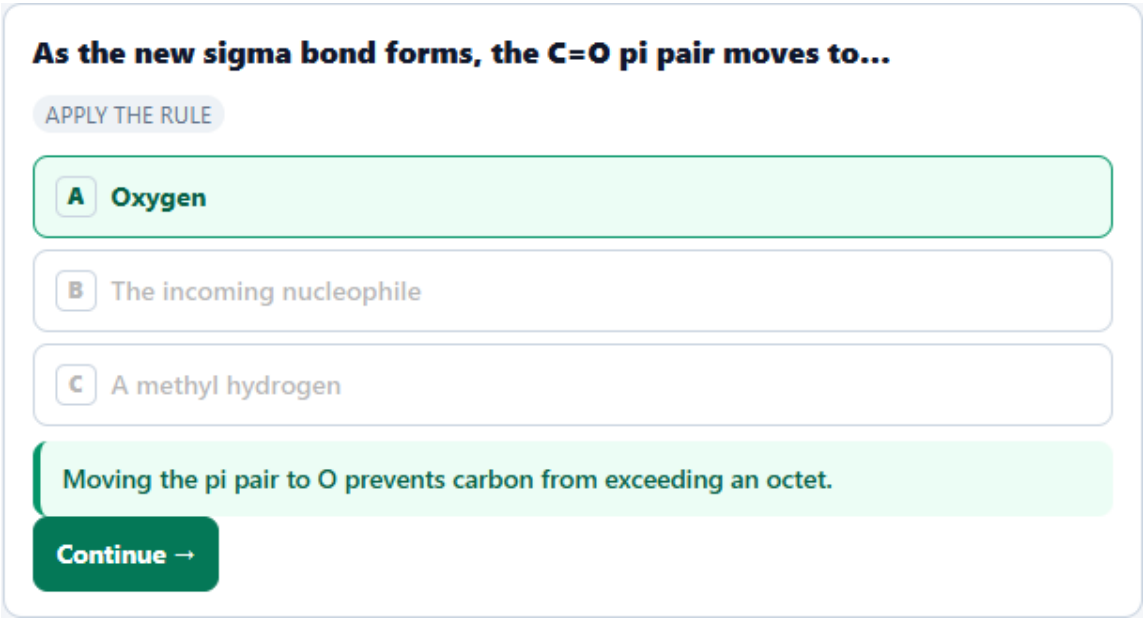
{"authorAnswer": 0, "tolerance": null, "highlightsAfterCorrectAnswer": {"flow": [{"fromBond": 1, "toAtom": 2}]}}

Hint 1: Count the carbon bonds during attack.

Hint 2: O can carry the resulting negative charge.

Explanation: Moving the pi pair to O prevents carbon from exceeding an octet.

Screenshot: addition-site-stage-2-feedback



addition-site:stage-3 | The immediate former-carbonyl geometry becomes approximately...

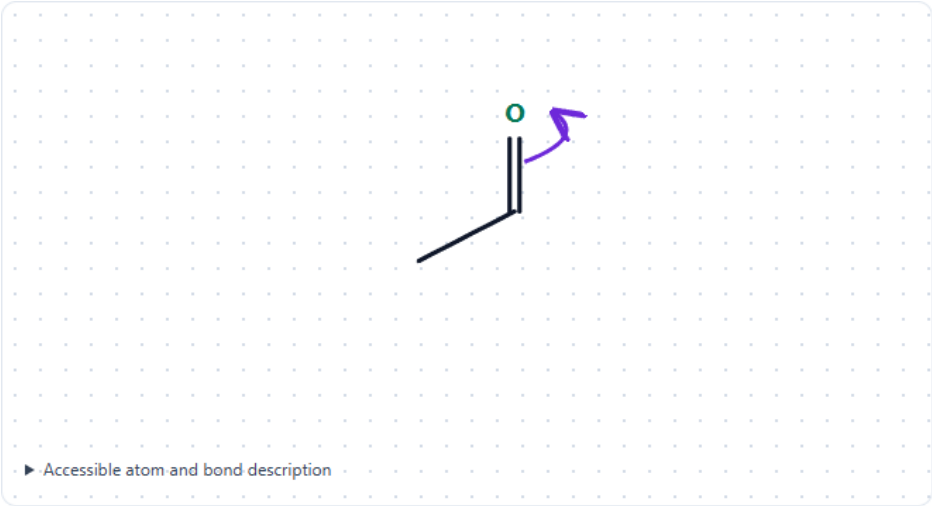
Answer type: choice

Option 0 (A): Tetrahedral

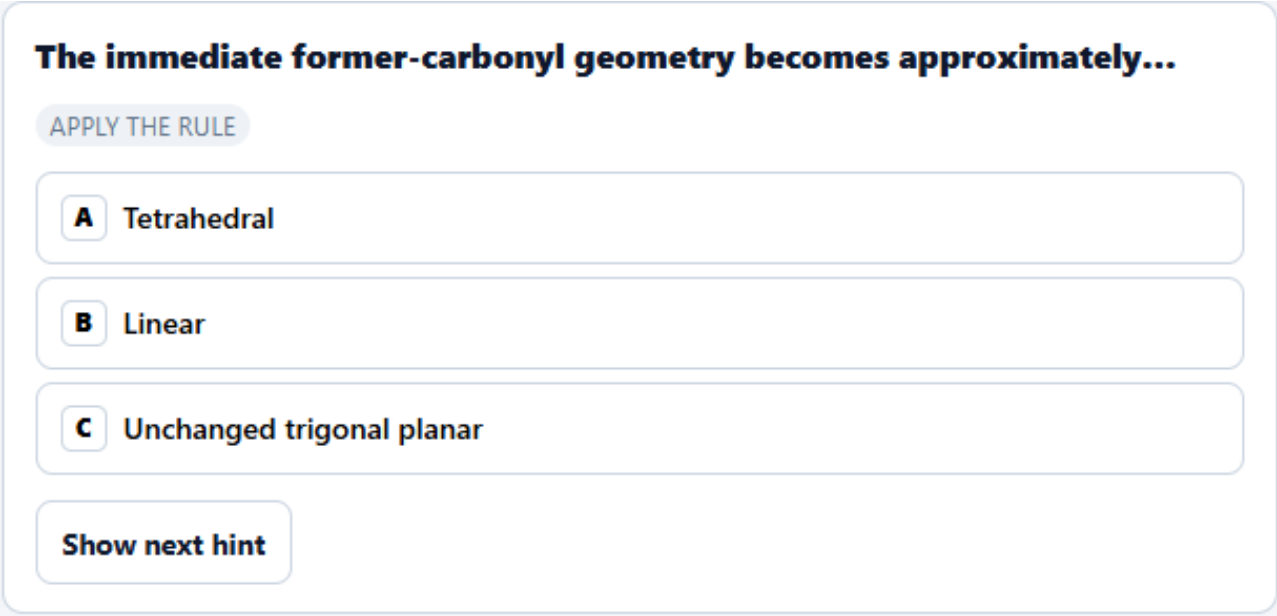
Option 1 (B): Linear

Option 2 (C): Unchanged trigonal planar

Screenshot: addition-site-stage-3-molecule



Screenshot: addition-site-stage-3-question



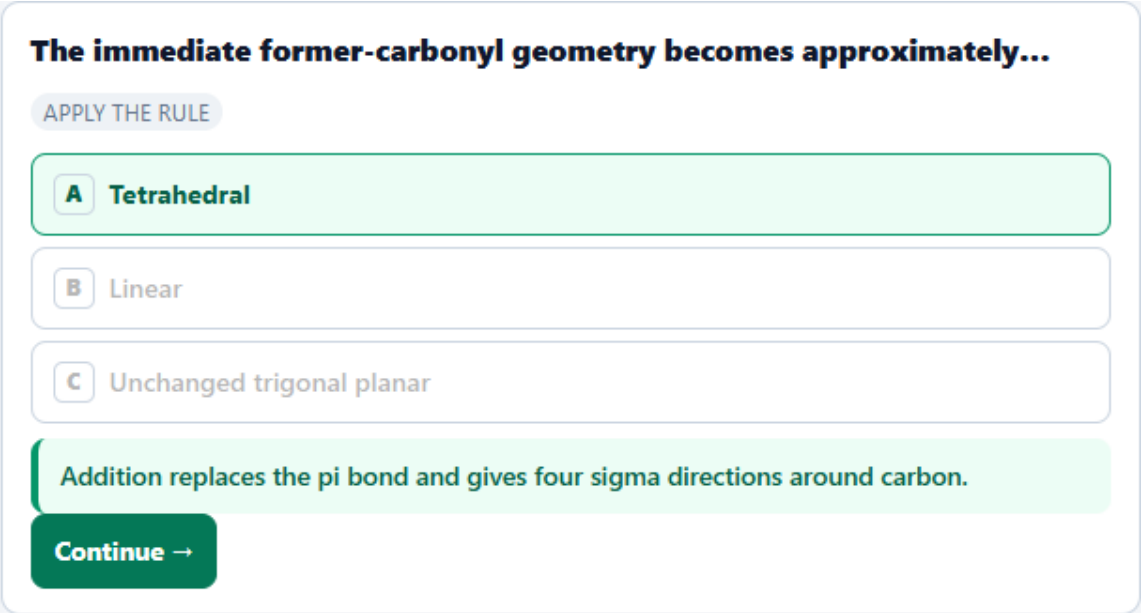
{"authorAnswer": 0, "tolerance": null, "highlightsAfterCorrectAnswer": {}}

Hint 1: Count electron domains after addition.

Hint 2: Four directions correspond to sp^3-like geometry.

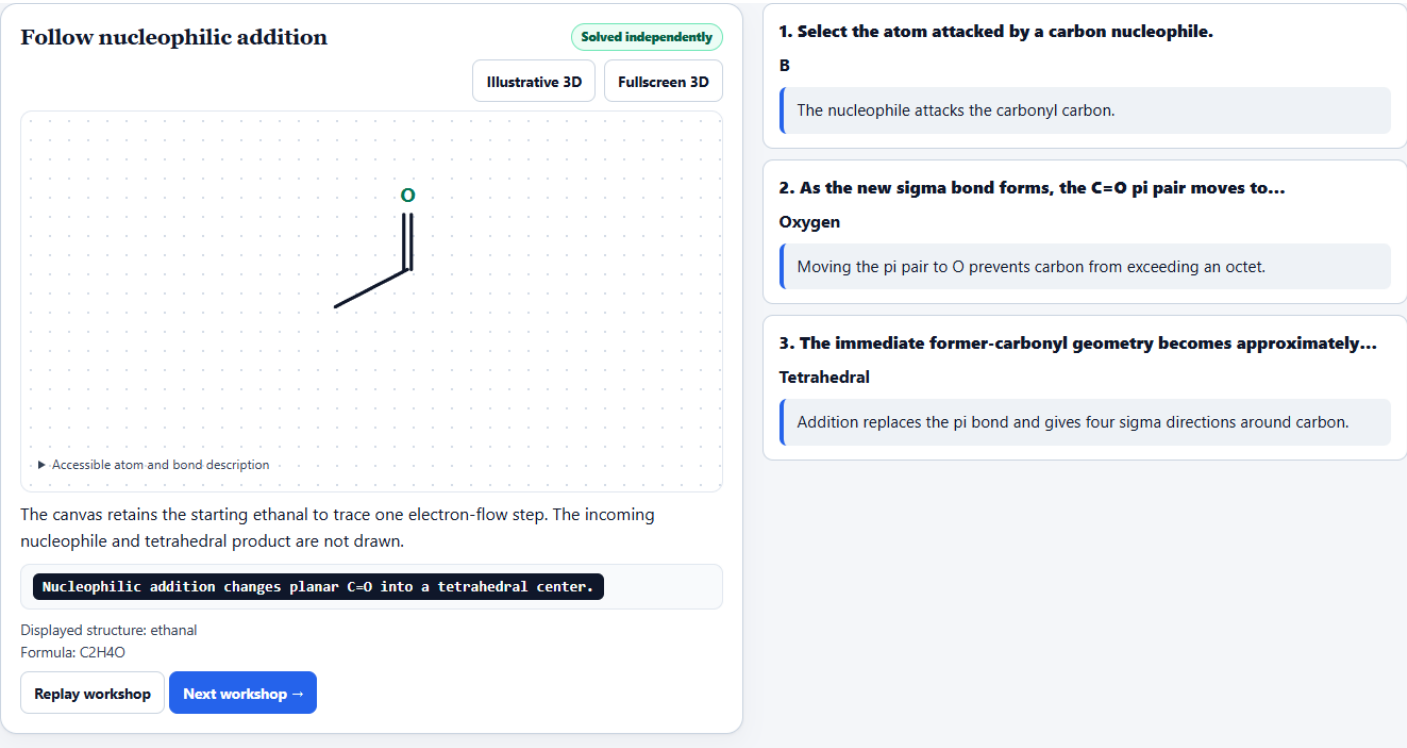
Explanation: Addition replaces the pi bond and gives four sigma directions around carbon.

Screenshot: addition-site-stage-3-feedback



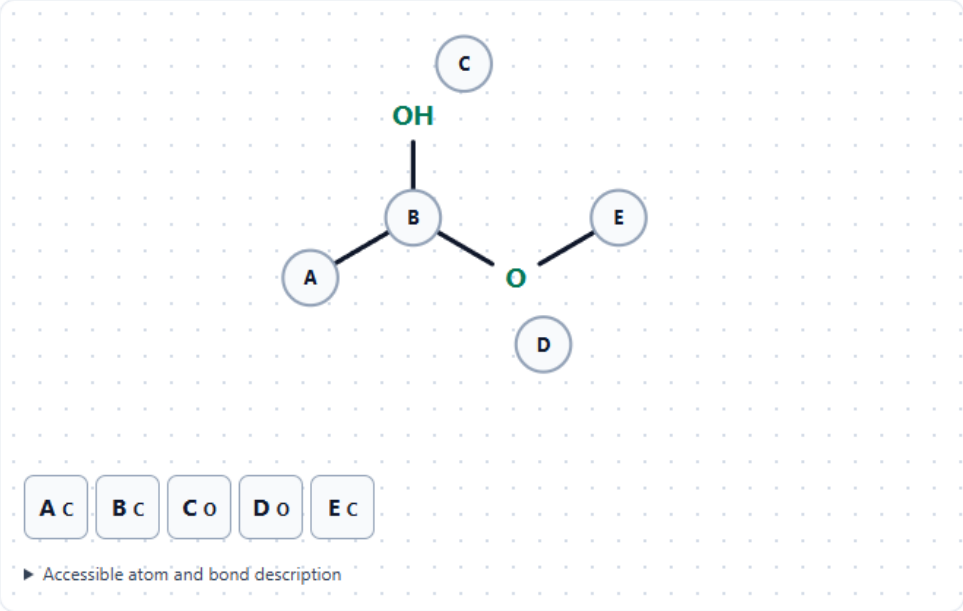
Completion summary: Nucleophilic addition changes planar C=O into a tetrahedral center.

Screenshot: addition-site-complete



hemiacetal-identify | Read oxygen substituents

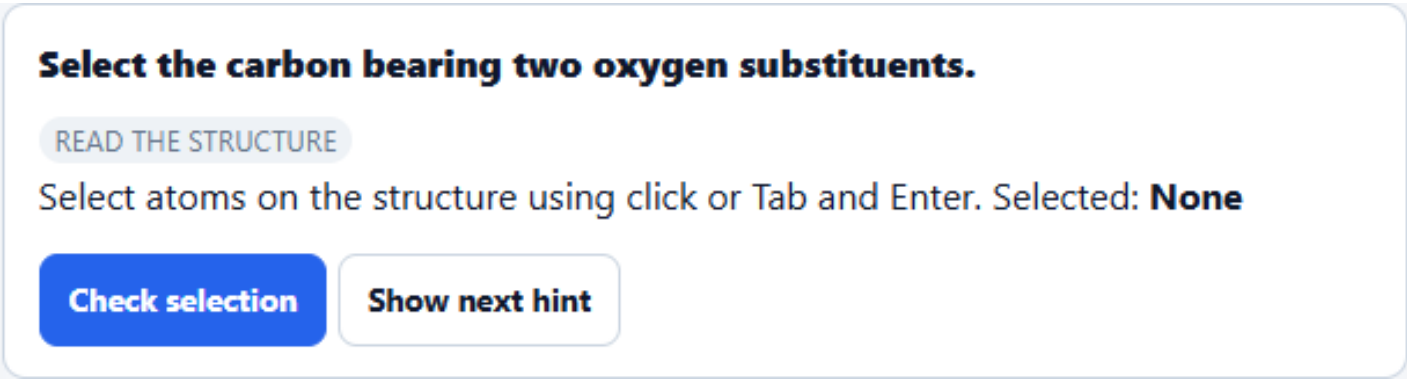
Screenshot: hemiacetal-identify-stage-1-molecule



hemiacetal-identify:stage-1 | Select the carbon bearing two oxygen substituents.

Answer type: atoms

Screenshot: hemiacetal-identify-stage-1-question



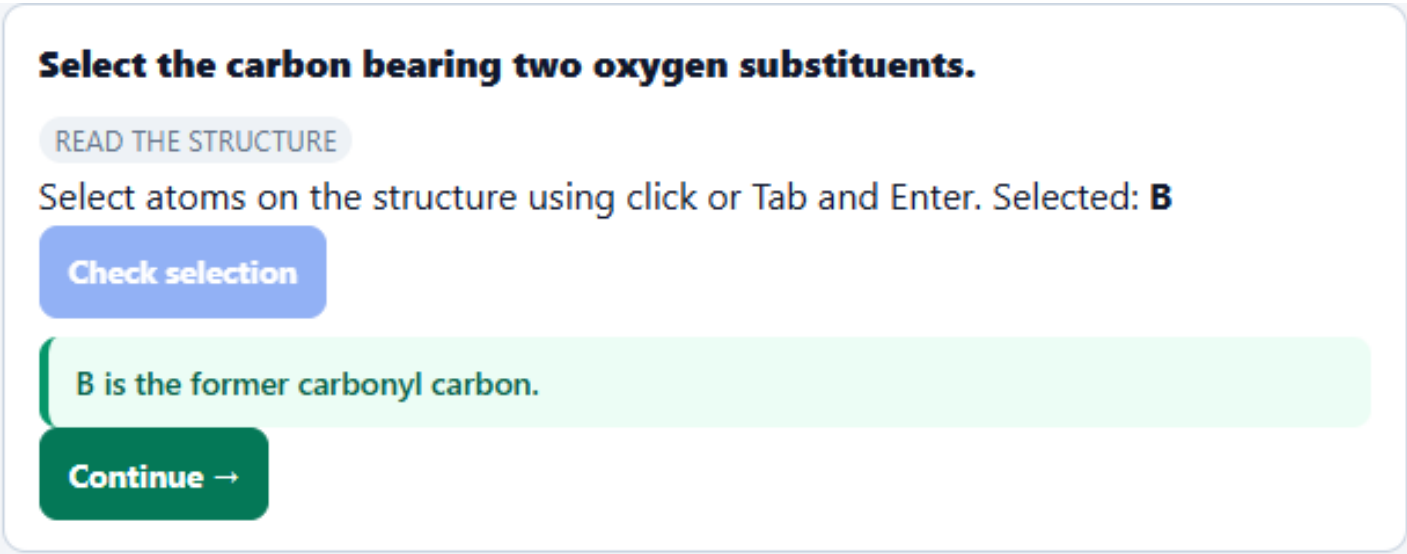
{"authorAnswer": [1], "tolerance": null, "highlightsAfterCorrectAnswer": {"prin": [1, 2, 3]}}

Hint 1: Trace both C–O bonds to their shared carbon.

Hint 2: The O atoms are C and D.

Explanation: B is the former carbonyl carbon.

Screenshot: hemiacetal-identify-stage-1-feedback



hemiacetal-identify:stage-2 | One OH and one OR on that carbon identify a...

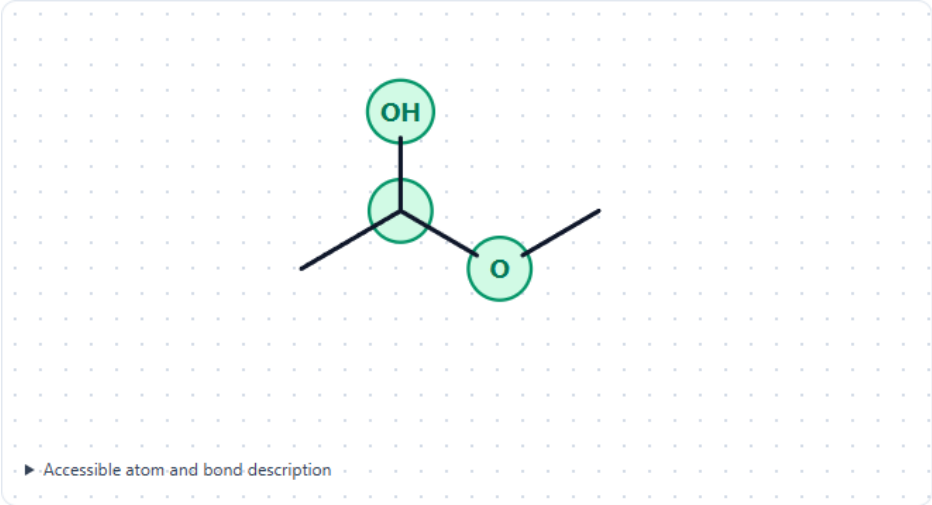
Answer type: choice

Option 0 (A): Hemiacetal

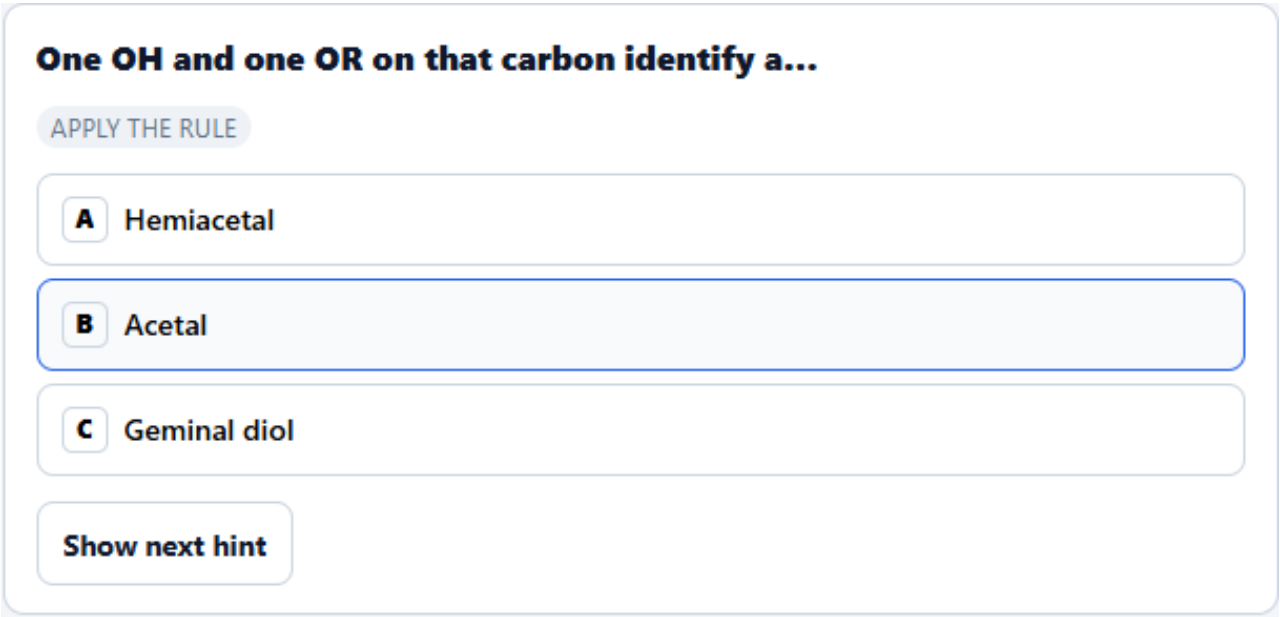
Option 1 (B): Acetal

Option 2 (C): Geminal diol

Screenshot: hemiacetal-identify-stage-2-molecule



Screenshot: hemiacetal-identify-stage-2-question



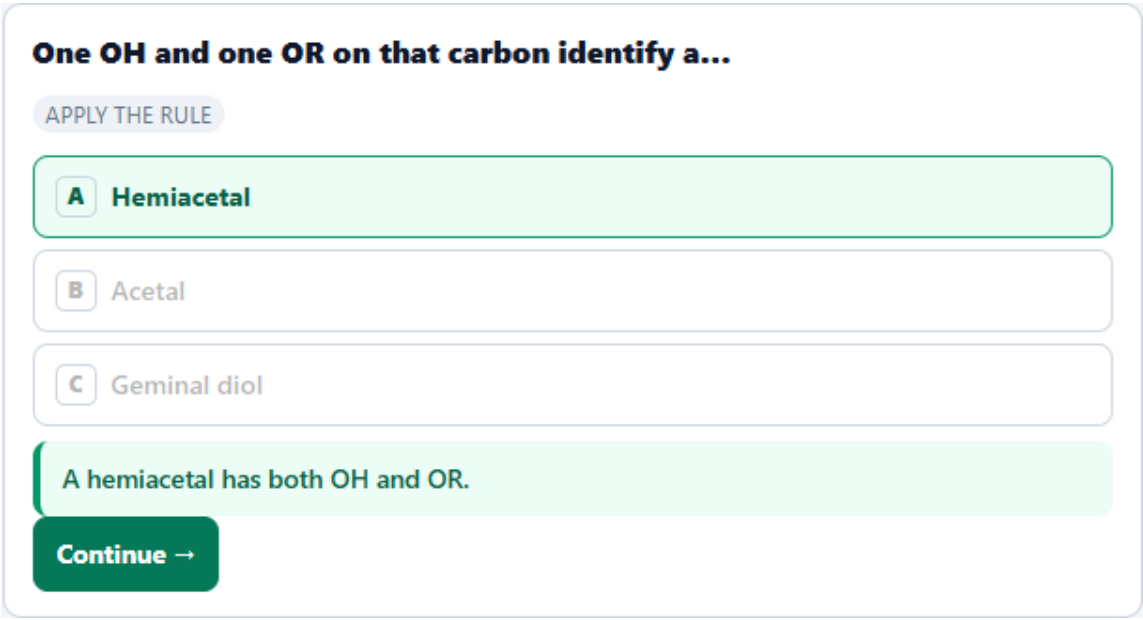
{"authorAnswer": 0, "tolerance": null, "highlightsAfterCorrectAnswer": {}}

Hint 1: Inspect whether each oxygen also has a carbon substituent.

Hint 2: One oxygen is OH; the other links to methyl.

Explanation: A hemiacetal has both OH and OR.

Screenshot: hemiacetal-identify-stage-2-feedback



hemiacetal-identify:stage-3 | Converting it to an acetal with alcohol is favored by...

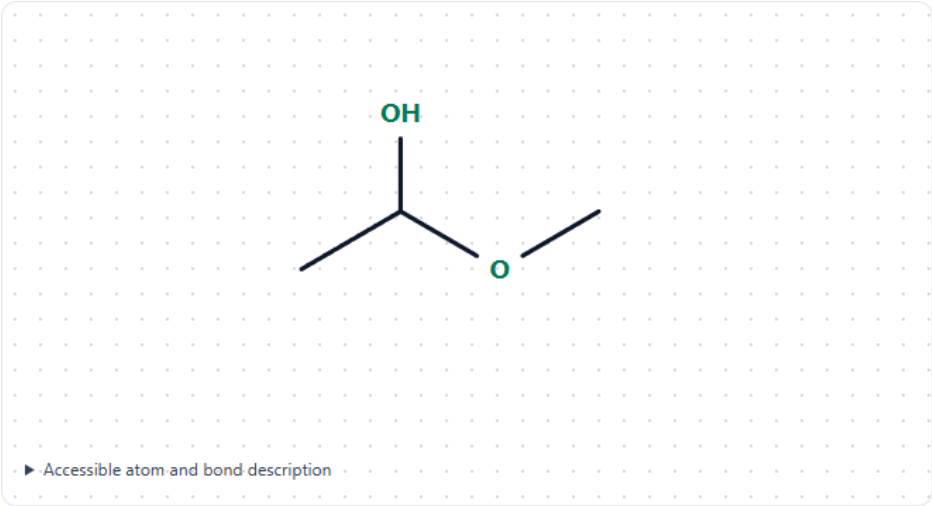
Answer type: choice

Option 0 (A): Acid catalysis and removal of water

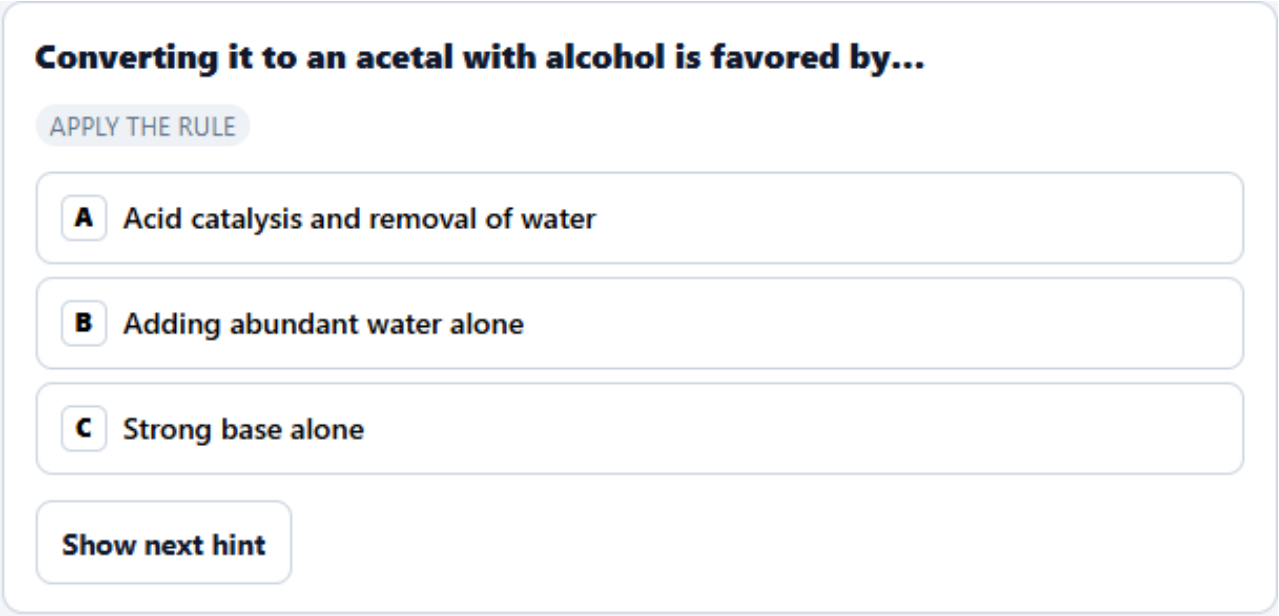
Option 1 (B): Adding abundant water alone

Option 2 (C): Strong base alone

Screenshot: hemiacetal-identify-stage-3-molecule



Screenshot: hemiacetal-identify-stage-3-question



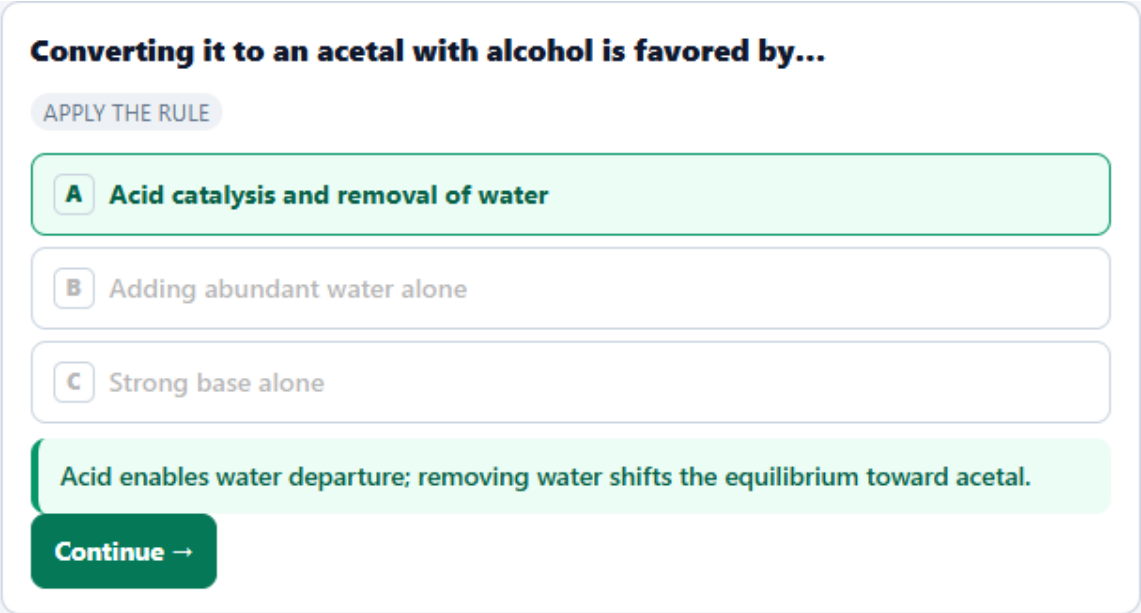
{"authorAnswer": 0, "tolerance": null, "highlightsAfterCorrectAnswer": {}}

Hint 1: Water is a product of acetal formation.

Hint 2: Apply Le Châtelier's principle.

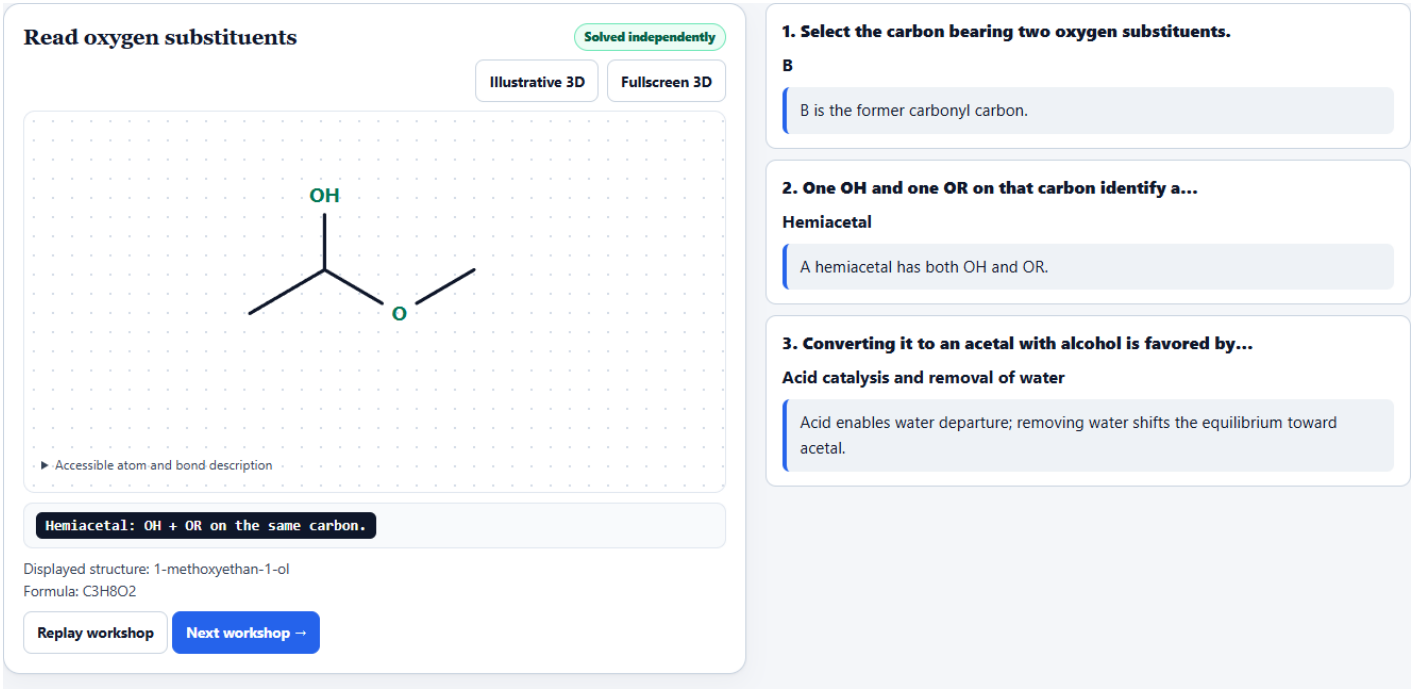
Explanation: Acid enables water departure; removing water shifts the equilibrium toward acetal.

Screenshot: hemiacetal-identify-stage-3-feedback



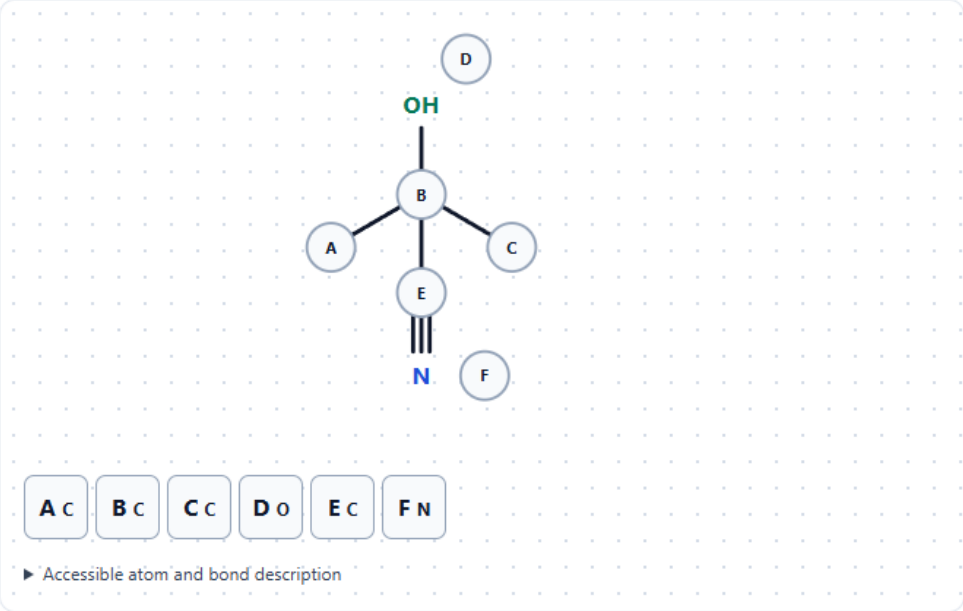
Completion summary: Hemiacetal: OH + OR on the same carbon.

Screenshot: hemiacetal-identify-complete



cyanohydrin-bond | Track the new carbon bond

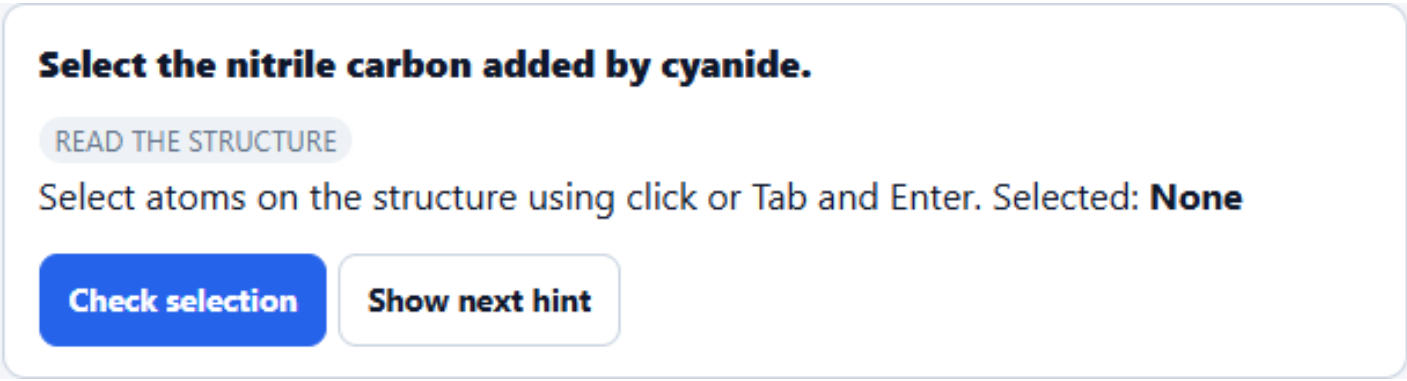
Screenshot: cyanohydrin-bond-stage-1-molecule



cyanohydrin-bond:stage-1 | Select the nitrile carbon added by cyanide.

Answer type: atoms

Screenshot: cyanohydrin-bond-stage-1-question



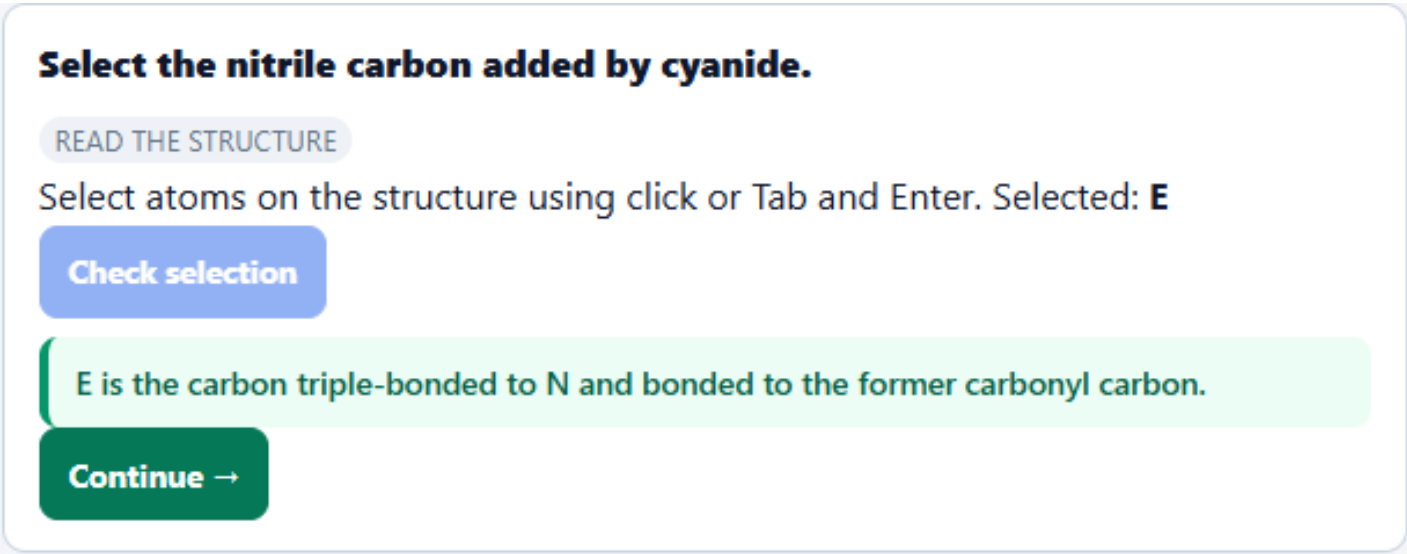
{"authorAnswer": [4], "tolerance": null, "highlightsAfterCorrectAnswer": {"prin": [4], "sub": [5]}}

Hint 1: Cyanide adds through carbon.

Hint 2: Find the carbon endpoint of C≡N.

Explanation: E is the carbon triple-bonded to N and bonded to the former carbonyl carbon.

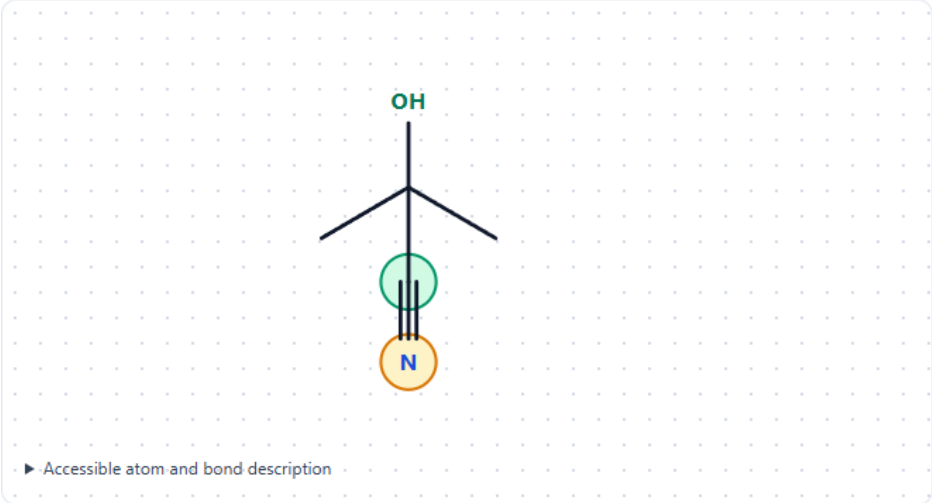
Screenshot: cyanohydrin-bond-stage-1-feedback



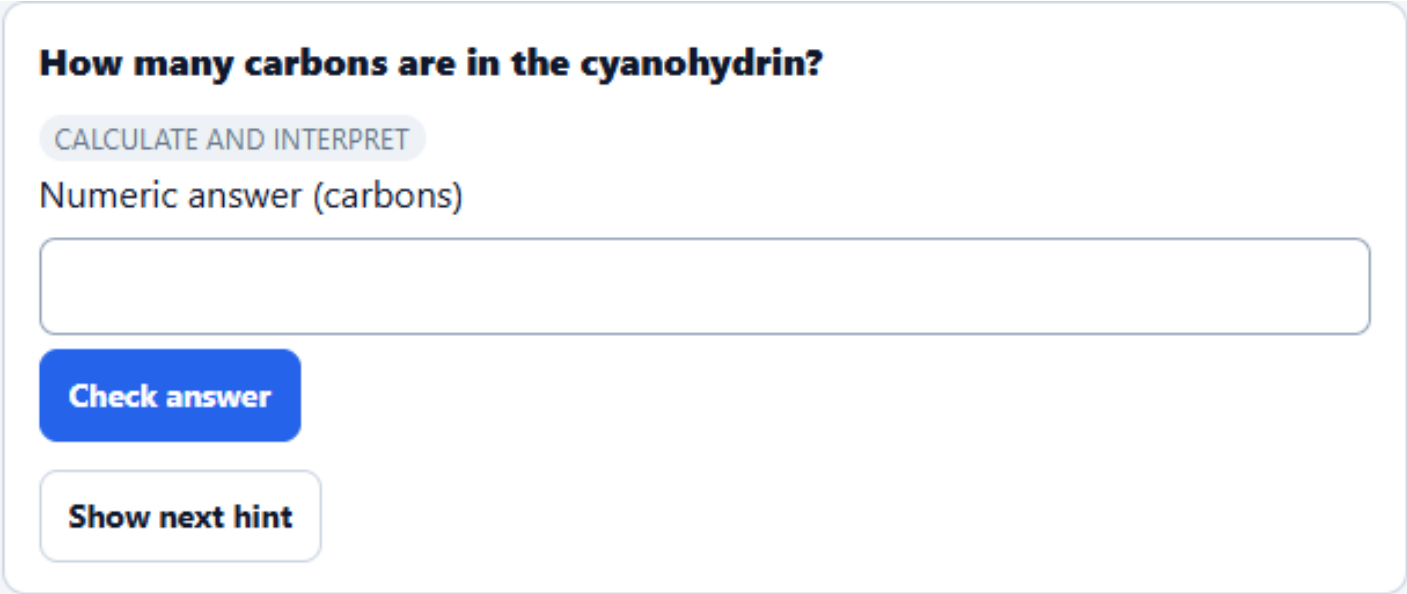
cyanohydrin-bond:stage-2 | How many carbons are in the cyanohydrin?

Answer type: number | unit: carbons

Screenshot: cyanohydrin-bond-stage-2-molecule



Screenshot: cyanohydrin-bond-stage-2-question



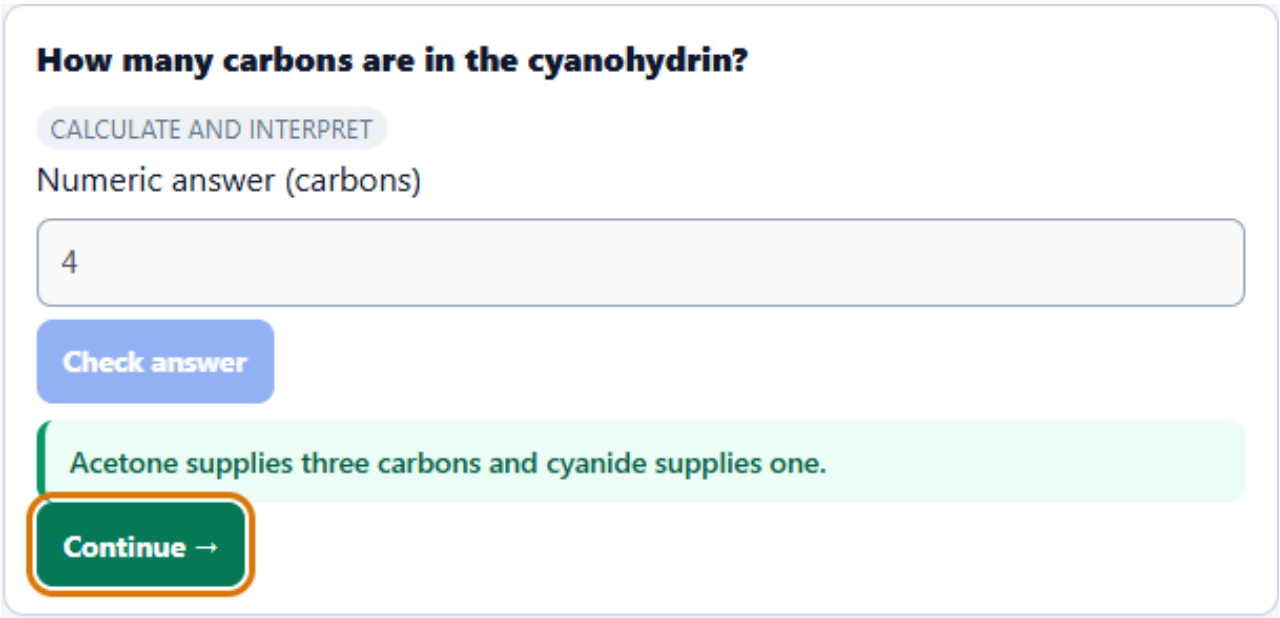
{"authorAnswer": 4, "tolerance": 0, "highlightsAfterCorrectAnswer": {}}

Hint 1: Count carbon atoms, not total atoms.

Hint 2: Include the nitrile carbon.

Explanation: Acetone supplies three carbons and cyanide supplies one.

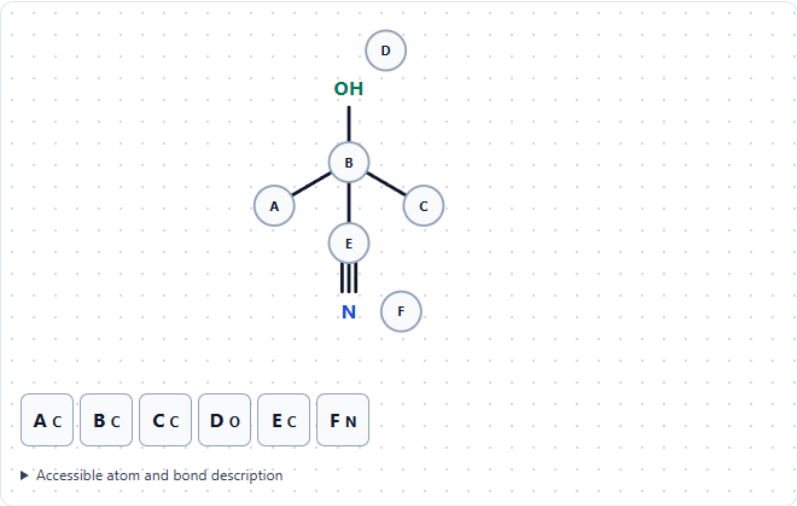
Screenshot: cyanohydrin-bond-stage-2-feedback



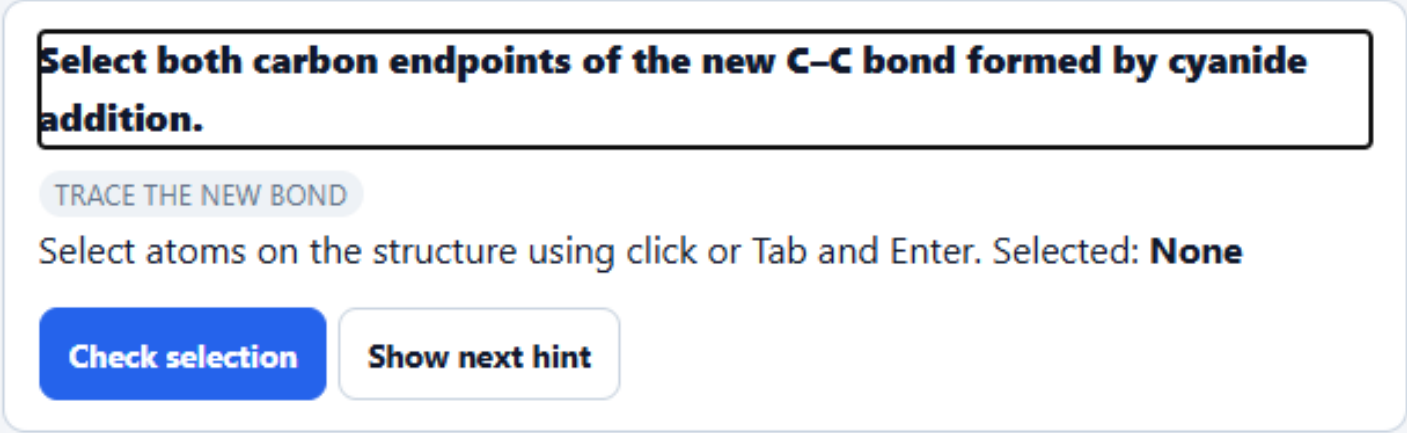
cyanohydrin-bond:stage-3 | Select both carbon endpoints of the new C–C bond formed by cyanide addition.

Answer type: atoms

Screenshot: cyanohydrin-bond-stage-3-molecule



Screenshot: cyanohydrin-bond-stage-3-question



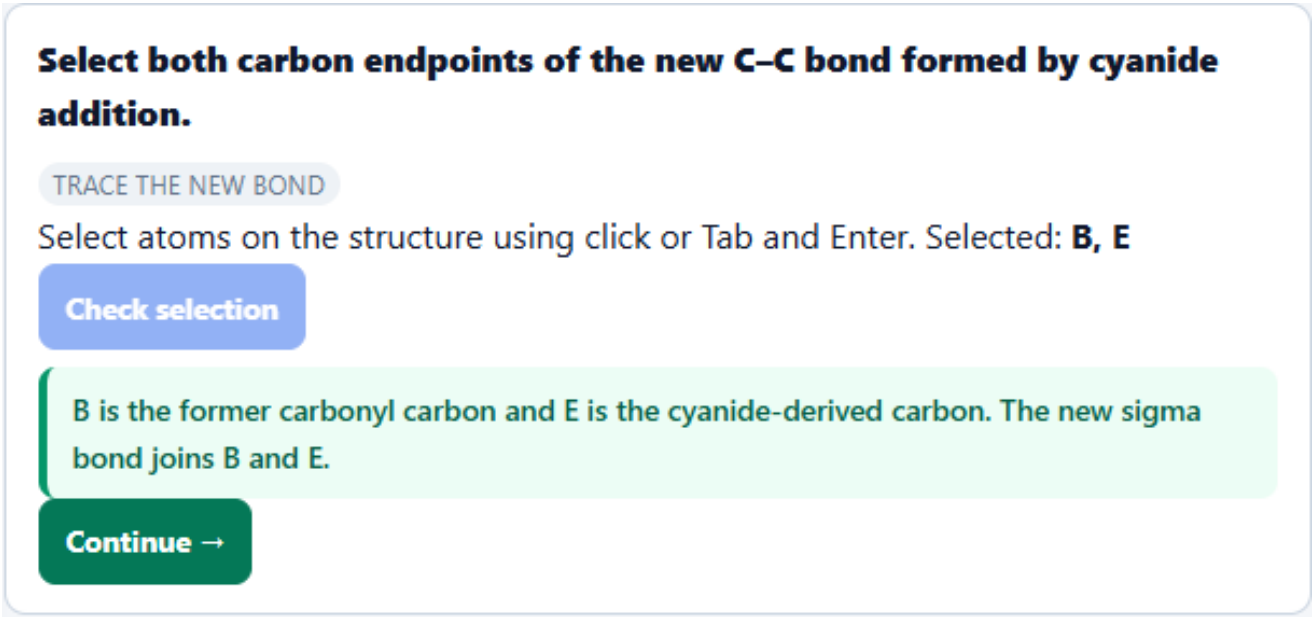
{"authorAnswer": [1, 4], "tolerance": null, "highlightsAfterCorrectAnswer": {"prin": [1, 4], "path": [1, 4]}}

Hint 1: Trace the single bond from the nitrile carbon toward the OH-bearing center.

Hint 2: Select two carbon atoms; the nitrogen endpoint of C≡N is not part of the new C–C bond.

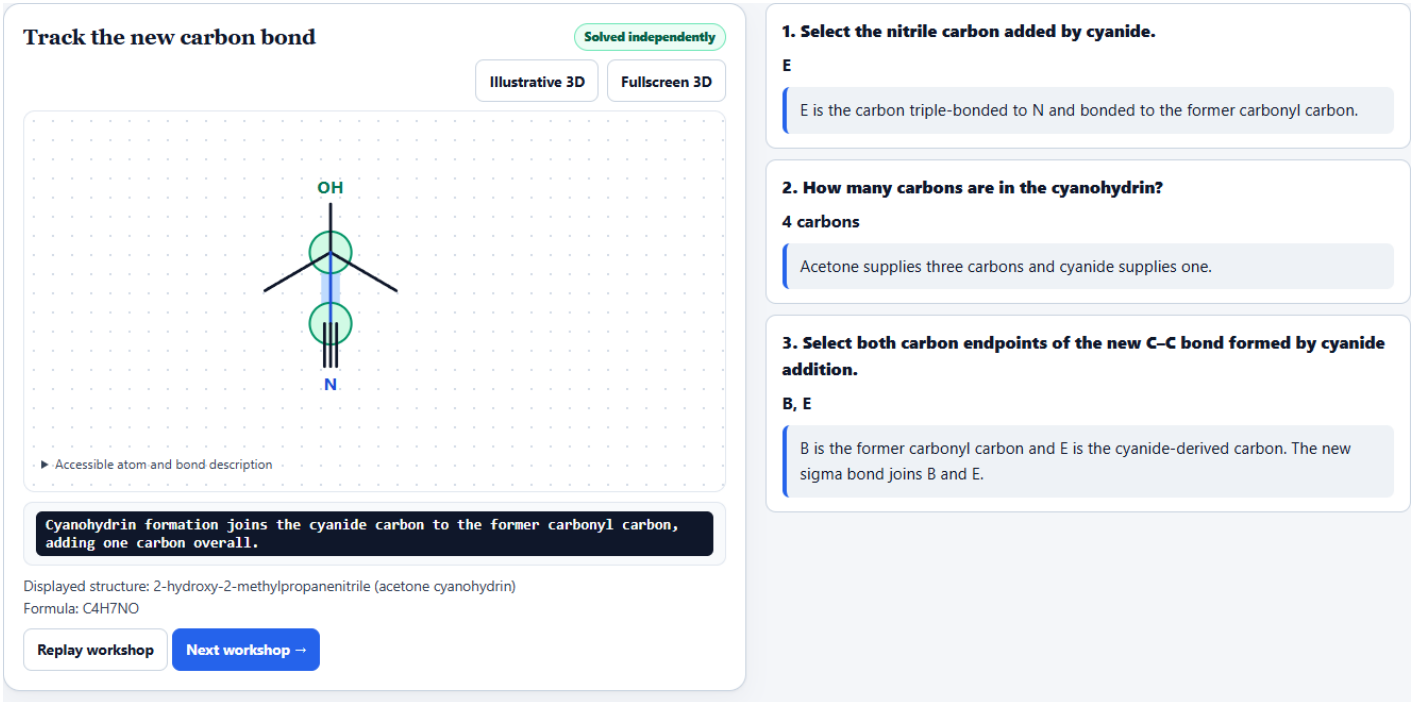
Explanation: B is the former carbonyl carbon and E is the cyanide-derived carbon. The new sigma bond joins B and E.

Screenshot: cyanohydrin-bond-stage-3-feedback



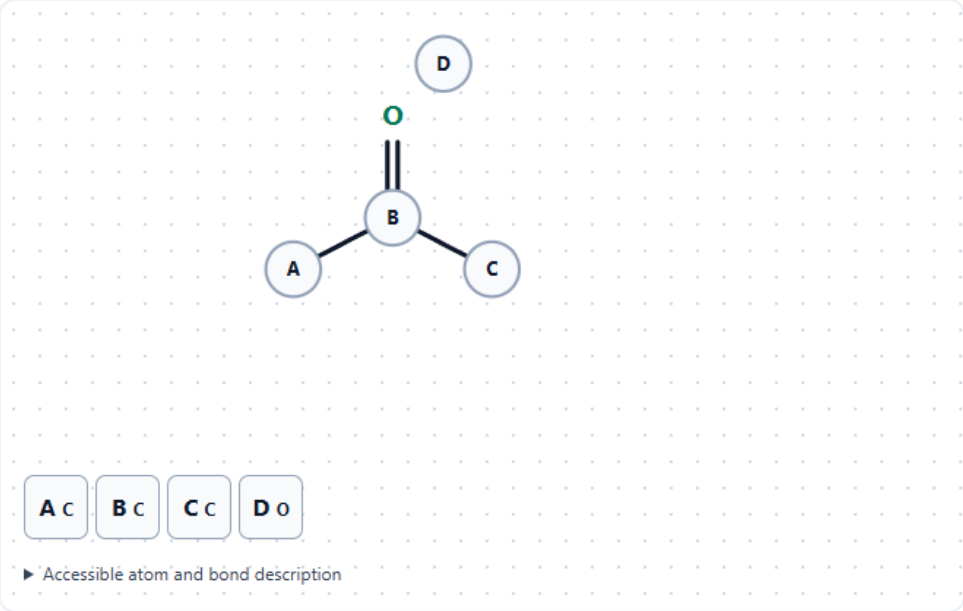
Completion summary: Cyanohydrin formation joins the cyanide carbon to the former carbonyl carbon, adding one carbon overall.

Screenshot: cyanohydrin-bond-complete



hydride-product | Predict carbonyl reduction

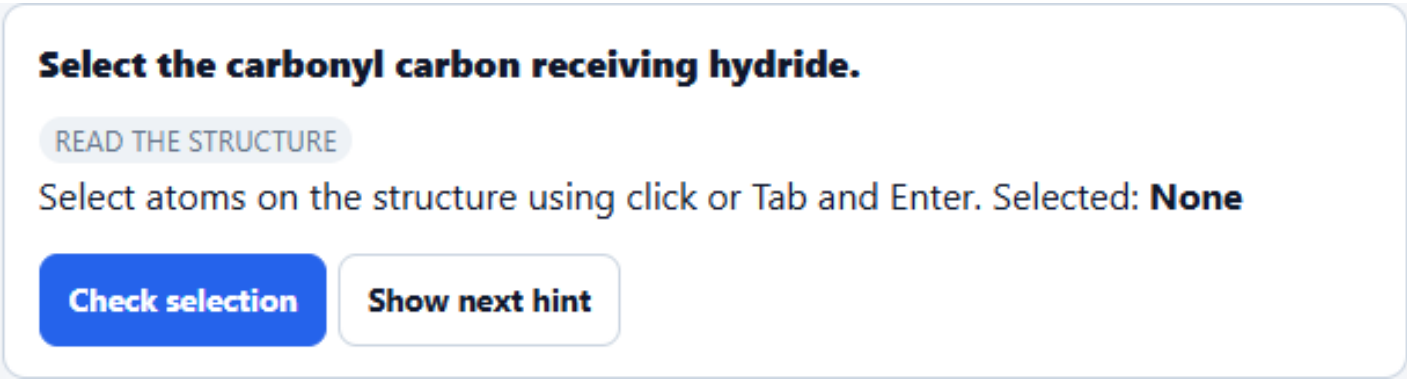
Screenshot: hydride-product-stage-1-molecule



hydride-product:stage-1 | Select the carbonyl carbon receiving hydride.

Answer type: atoms

Screenshot: hydride-product-stage-1-question



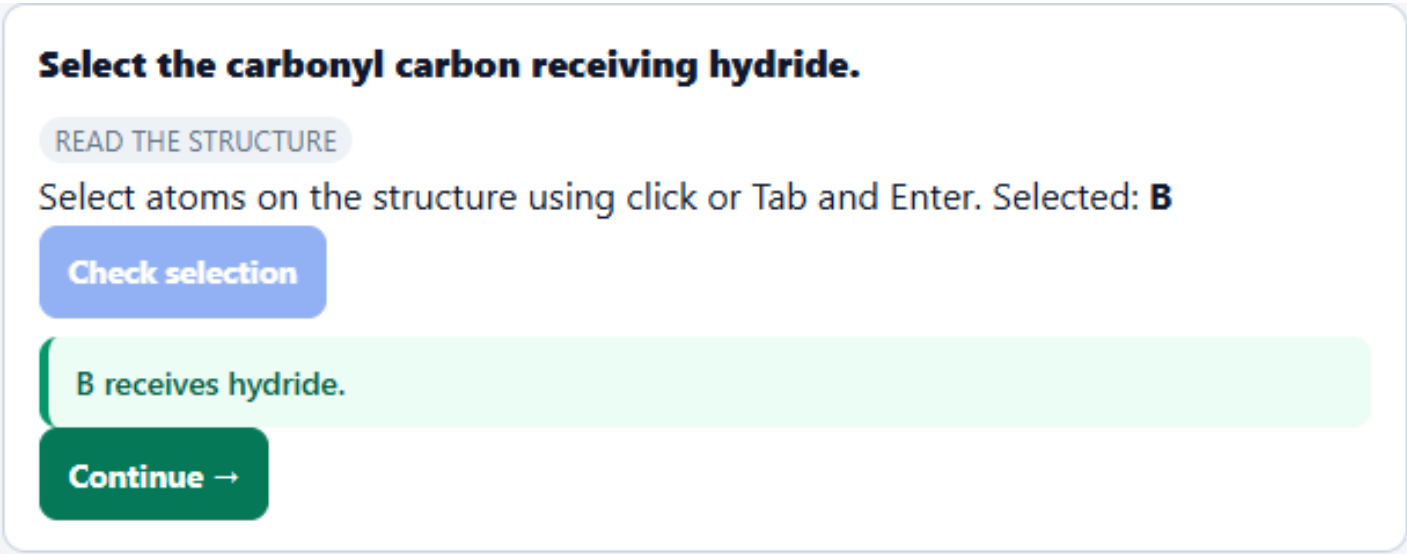
{"authorAnswer": [1], "tolerance": null, "highlightsAfterCorrectAnswer": {"prin": [1, 3]}}

Hint 1: Hydride is the nucleophile.

Hint 2: Attack the electron-poor carbon.

Explanation: B receives hydride.

Screenshot: hydride-product-stage-1-feedback



hydride-product:stage-2 | After protonation, the product is a...

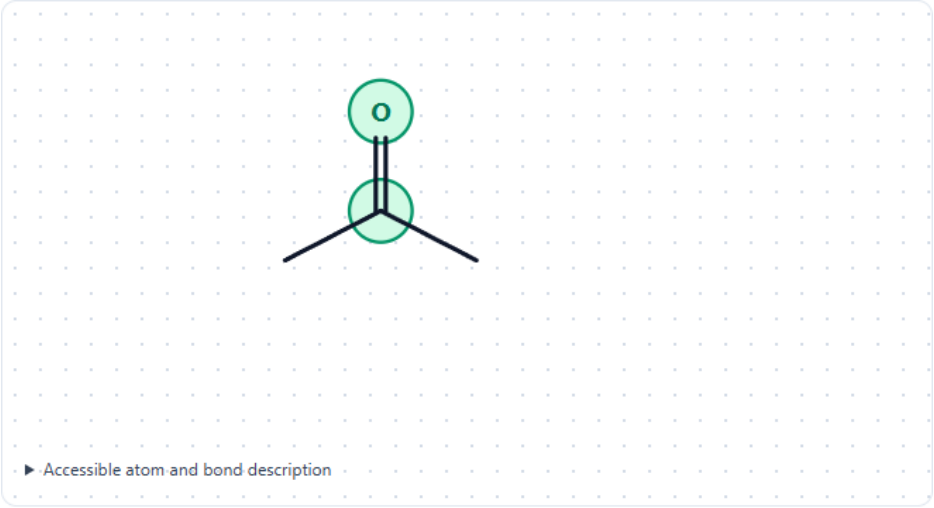
Answer type: choice

Option 0 (A): Secondary alcohol

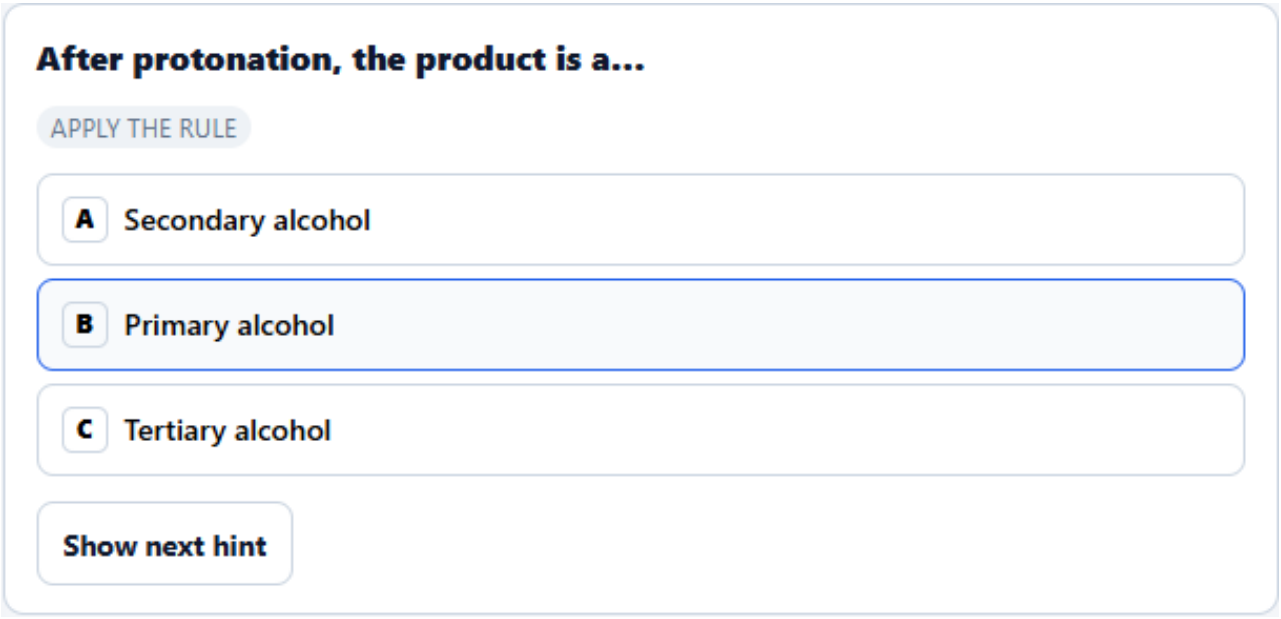
Option 1 (B): Primary alcohol

Option 2 (C): Tertiary alcohol

Screenshot: hydride-product-stage-2-molecule



Screenshot: hydride-product-stage-2-question



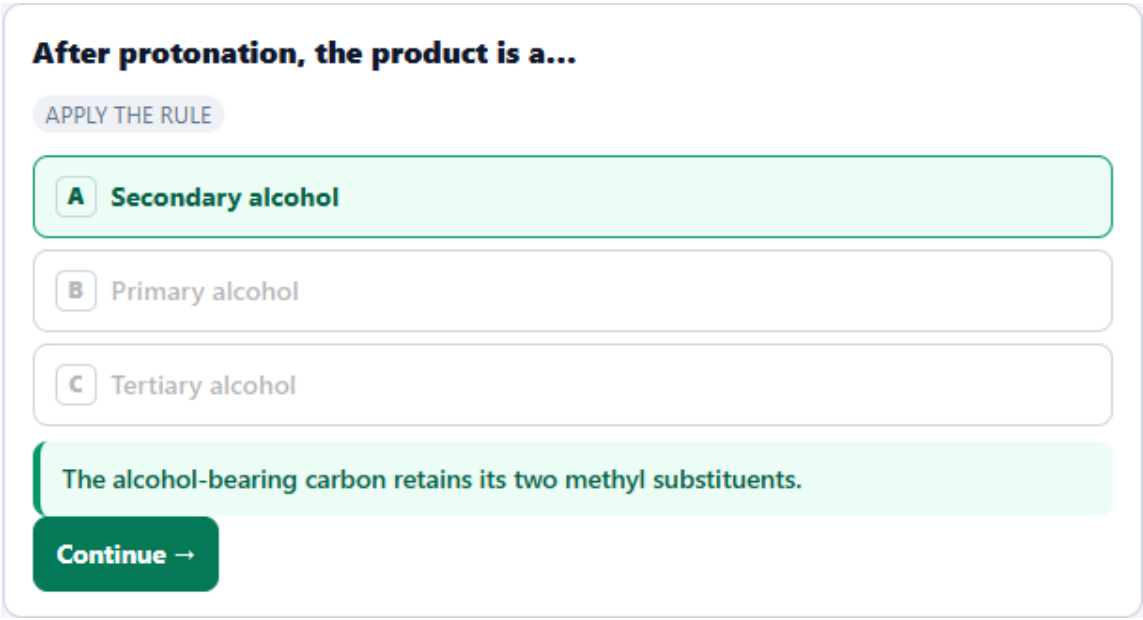
{"authorAnswer": 0, "tolerance": null, "highlightsAfterCorrectAnswer": {}}

Hint 1: Count carbon neighbors of the new C–OH carbon.

Hint 2: No C–C bonds are added or removed.

Explanation: The alcohol-bearing carbon retains its two methyl substituents.

Screenshot: hydride-product-stage-2-feedback

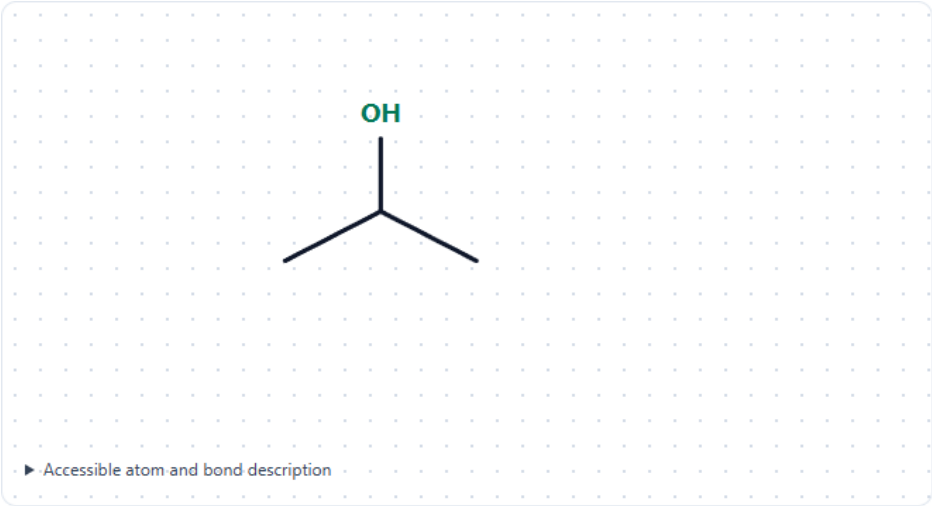


This stage uses a different molecule (connectivity shown).

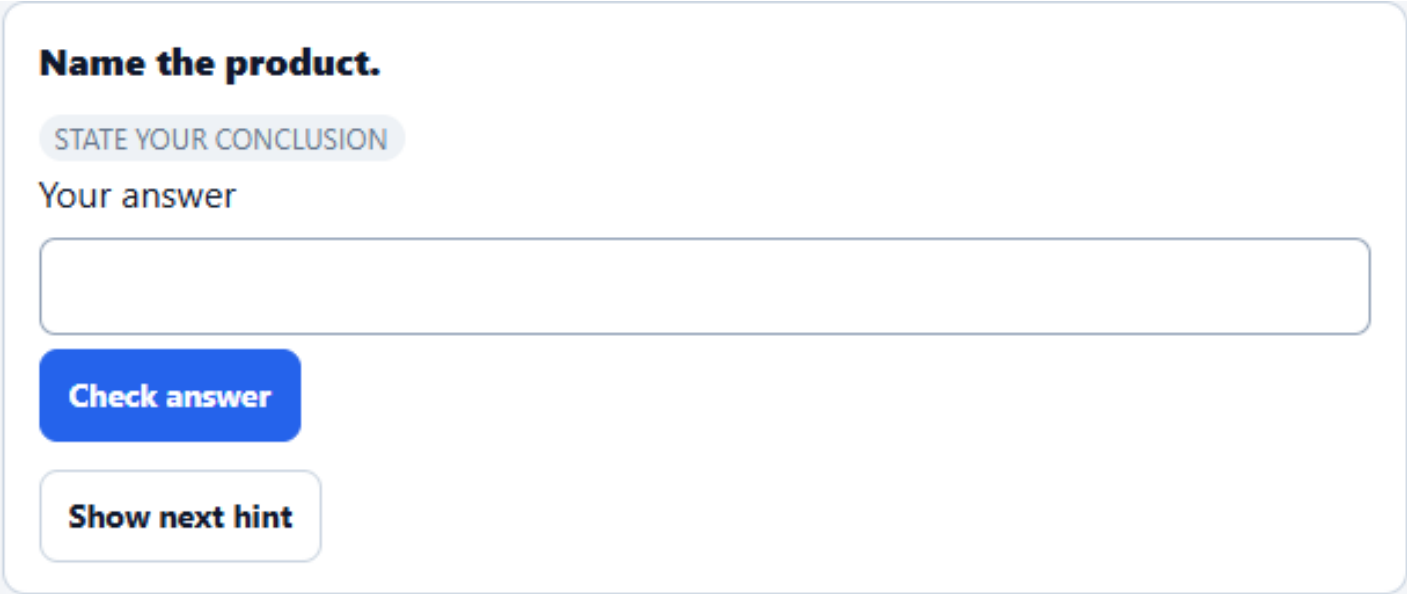
hydride-product:stage-3 | Name the product.

Answer type: text

Screenshot: hydride-product-stage-3-molecule



Screenshot: hydride-product-stage-3-question



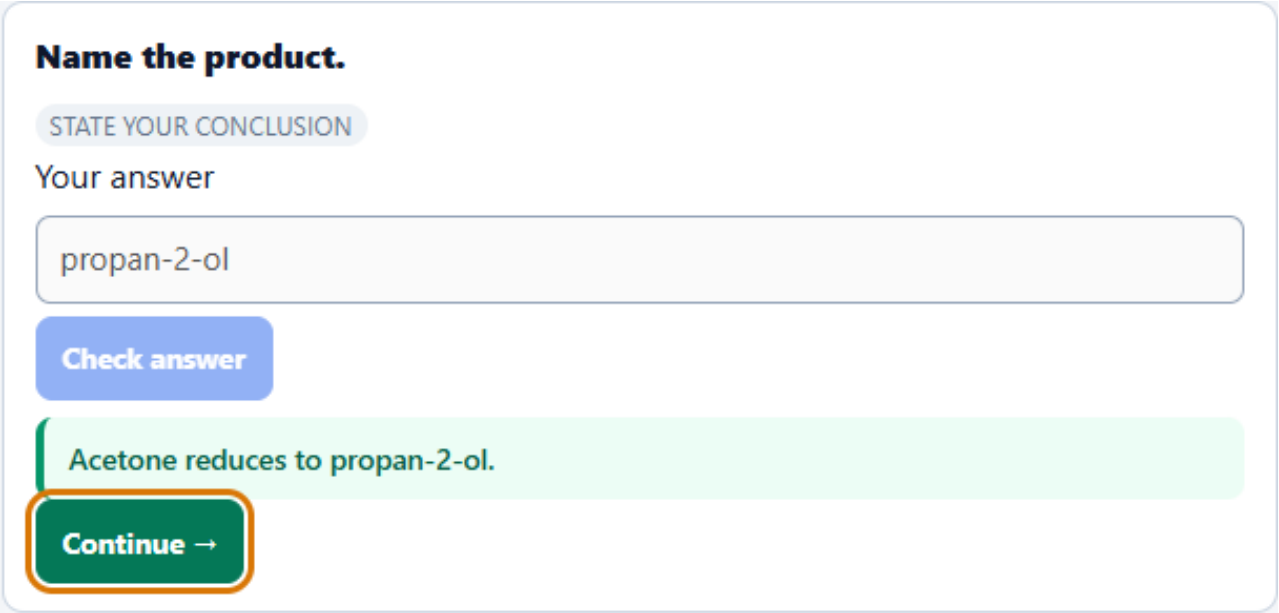
{"authorAnswer": ["propan-2-ol", "2-propanol", "isopropanol", "isopropyl alcohol"], "tolerance": null, "highlightsAfterCorrectAnswer": {}}

Hint 1: Retain the three-carbon chain.

Hint 2: Place OH at the central carbon.

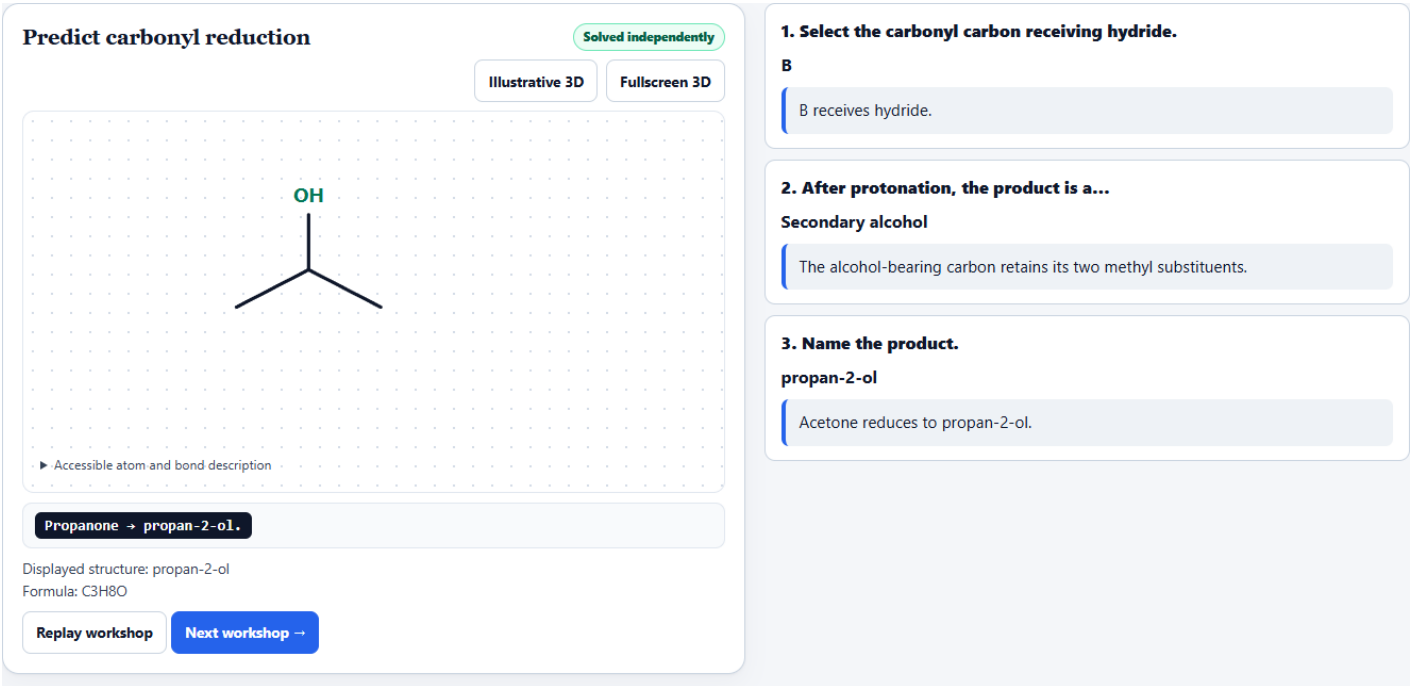
Explanation: Acetone reduces to propan-2-ol.

Screenshot: hydride-product-stage-3-feedback



Completion summary: Propanone → propan-2-ol.

Screenshot: hydride-product-complete



Molecular repository: author claims to verify

acetaldehyde

Author claims: ethanal | formula C2H4O | canonical composition C2H4O | net charge +0

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=131.00, y=115.00; charge=0; explicit H override=none

1 (B): C; x=189.00, y=85.00; charge=0; explicit H override=none

2 (C): O; x=189.00, y=25.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 2]]

hemiacetal

Author claims: 1-methoxyethan-1-ol | formula C3H8O2 | canonical composition C3H8O2 | net charge +0

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=60.00, y=120.00; charge=0; explicit H override=none

1 (B): C; x=120.00, y=85.00; charge=0; explicit H override=none

2 (C): O; x=120.00, y=25.00; charge=0; explicit H override=none

3 (D): O; x=180.00, y=120.00; charge=0; explicit H override=none

4 (E): C; x=240.00, y=85.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 1], [1, 3, 1], [3, 4, 1]]

acetal

Author claims: 2,2-dimethoxypropane | formula C5H12O2 | canonical composition C5H12O2 | net charge +0

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=75.00, y=130.00; charge=0; explicit H override=none

1 (B): C; x=135.00, y=100.00; charge=0; explicit H override=none

2 (C): C; x=195.00, y=130.00; charge=0; explicit H override=none

3 (D): O; x=105.00, y=45.00; charge=0; explicit H override=none

4 (E): O; x=165.00, y=45.00; charge=0; explicit H override=none

5 (F): C; x=50.00, y=15.00; charge=0; explicit H override=none

6 (G): C; x=220.00, y=15.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 1], [1, 3, 1], [1, 4, 1], [3, 5, 1], [4, 6, 1]]

geminal

Author claims: propane-2,2-diol | formula C3H8O2 | canonical composition C3H8O2 | net charge +0

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=48.00, y=115.00; charge=0; explicit H override=none

1 (B): C; x=102.00, y=85.00; charge=0; explicit H override=none

2 (C): C; x=156.00, y=115.00; charge=0; explicit H override=none

3 (D): O; x=75.00, y=32.00; charge=0; explicit H override=none

4 (E): O; x=129.00, y=32.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 1], [1, 3, 1], [1, 4, 1]]

imine

Author claims: N-methylethanimine (E/Z unspecified) | formula C3H7N | canonical composition C3H7N | net charge +0

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=55.00, y=120.00; charge=0; explicit H override=none

1 (B): C; x=115.00, y=85.00; charge=0; explicit H override=none

2 (C): N; x=180.00, y=85.00; charge=0; explicit H override=none

3 (D): C; x=240.00, y=120.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 2], [2, 3, 1]]

enamine

Author claims: N,N-dimethylethenamine | formula C4H9N | canonical composition C4H9N | net charge +0

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=55.00, y=120.00; charge=0; explicit H override=none

1 (B): C; x=115.00, y=85.00; charge=0; explicit H override=none

2 (C): N; x=180.00, y=120.00; charge=0; explicit H override=none

3 (D): C; x=240.00, y=85.00; charge=0; explicit H override=none

4 (E): C; x=180.00, y=180.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 2], [1, 2, 1], [2, 3, 1], [2, 4, 1]]

cyanohydrin

Author claims: 2-hydroxy-2-methylpropanenitrile (acetone cyanohydrin) | formula C4H7NO | canonical composition C4H7NO | net charge +0

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=60.00, y=120.00; charge=0; explicit H override=none

1 (B): C; x=120.00, y=85.00; charge=0; explicit H override=none

2 (C): C; x=180.00, y=120.00; charge=0; explicit H override=none

3 (D): O; x=120.00, y=25.00; charge=0; explicit H override=none

4 (E): C; x=120.00, y=150.00; charge=0; explicit H override=none

5 (F): N; x=120.00, y=205.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 1], [1, 3, 1], [1, 4, 1], [4, 5, 3]]

aceticacid

Author claims: ethanoic acid | formula C2H4O2 | canonical composition C2H4O2 | net charge +0

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=50.00, y=115.00; charge=0; explicit H override=none

1 (B): C; x=108.00, y=85.00; charge=0; explicit H override=none

2 (C): O; x=108.00, y=25.00; charge=0; explicit H override=none

3 (D): O; x=166.00, y=115.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 2], [1, 3, 1]]

ethanol

Author claims: ethanol | formula C2H6O | canonical composition C2H6O | net charge +0

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=50.00, y=115.00; charge=0; explicit H override=none

1 (B): C; x=108.00, y=85.00; charge=0; explicit H override=none

2 (C): O; x=166.00, y=115.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 1]]

acetone

Author claims: propanone | formula C3H6O | canonical composition C3H6O | net charge +0

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=50.00, y=115.00; charge=0; explicit H override=none

1 (B): C; x=108.00, y=85.00; charge=0; explicit H override=none

2 (C): C; x=166.00, y=115.00; charge=0; explicit H override=none

3 (D): O; x=108.00, y=25.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 1], [1, 3, 2]]

isopropanol

Author claims: propan-2-ol | formula C3H8O | canonical composition C3H8O | net charge +0

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=50.00, y=115.00; charge=0; explicit H override=none

1 (B): C; x=108.00, y=85.00; charge=0; explicit H override=none

2 (C): C; x=166.00, y=115.00; charge=0; explicit H override=none

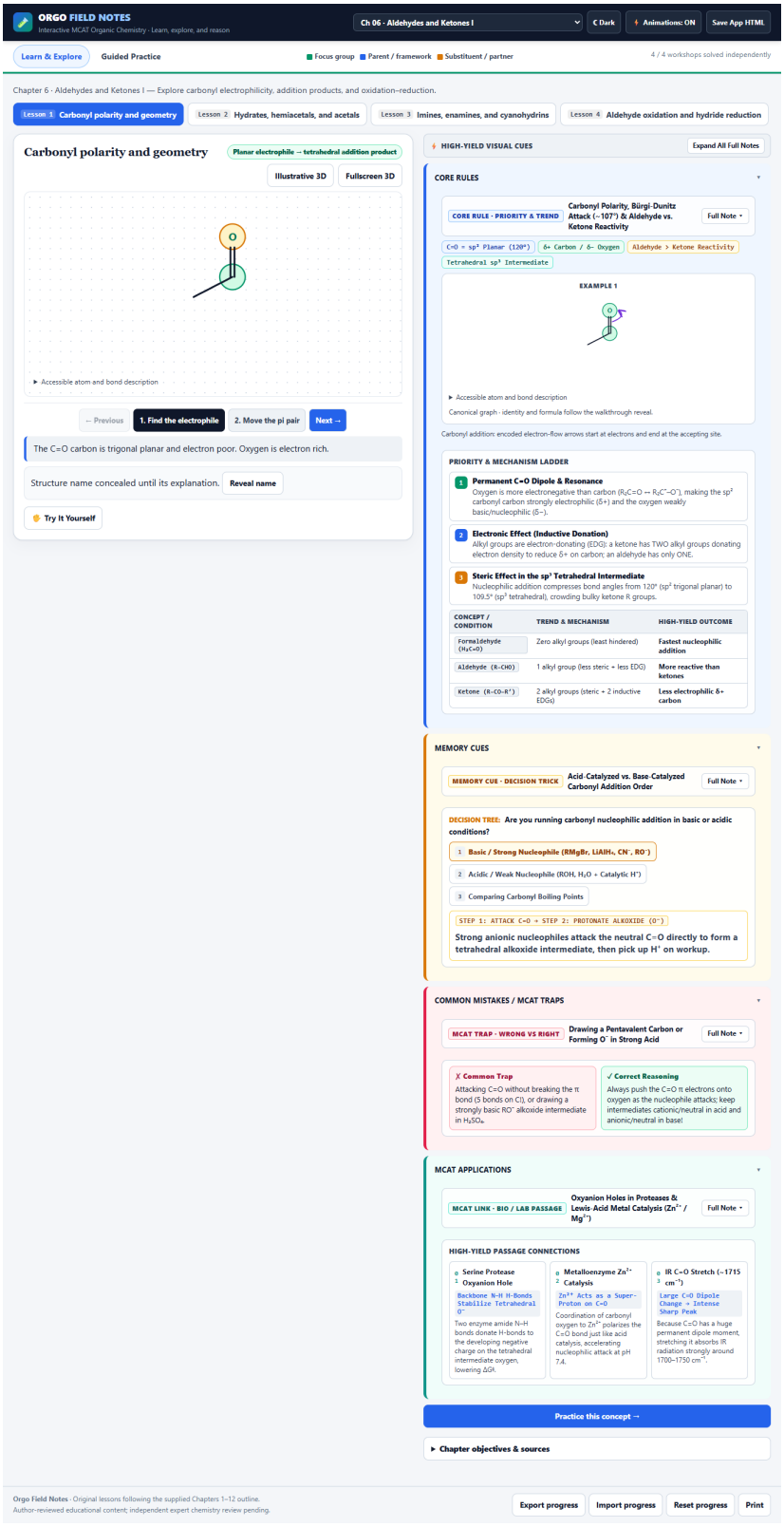
3 (D): O; x=108.00, y=25.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 1], [1, 3, 1]]

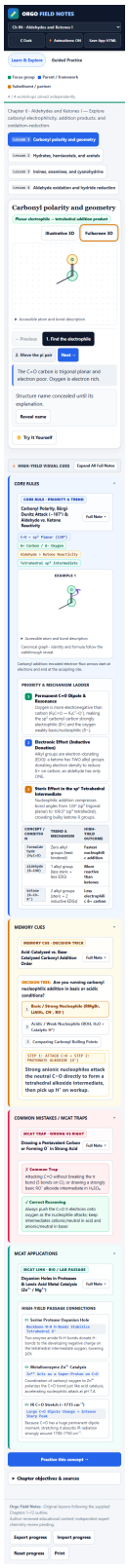
Layout samples

These overview images are scaled for page fit. Detailed structure and stage images appear earlier. Samples cover the first concept on desktop and mobile in both themes, not every responsive state.

Screenshot: layout-desktop-light



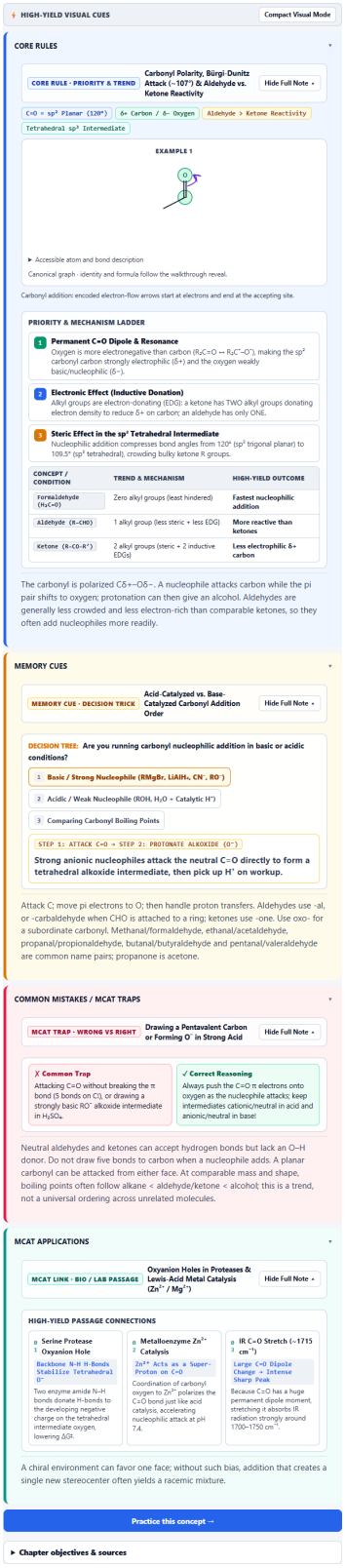
Screenshot: layout-mobile-light



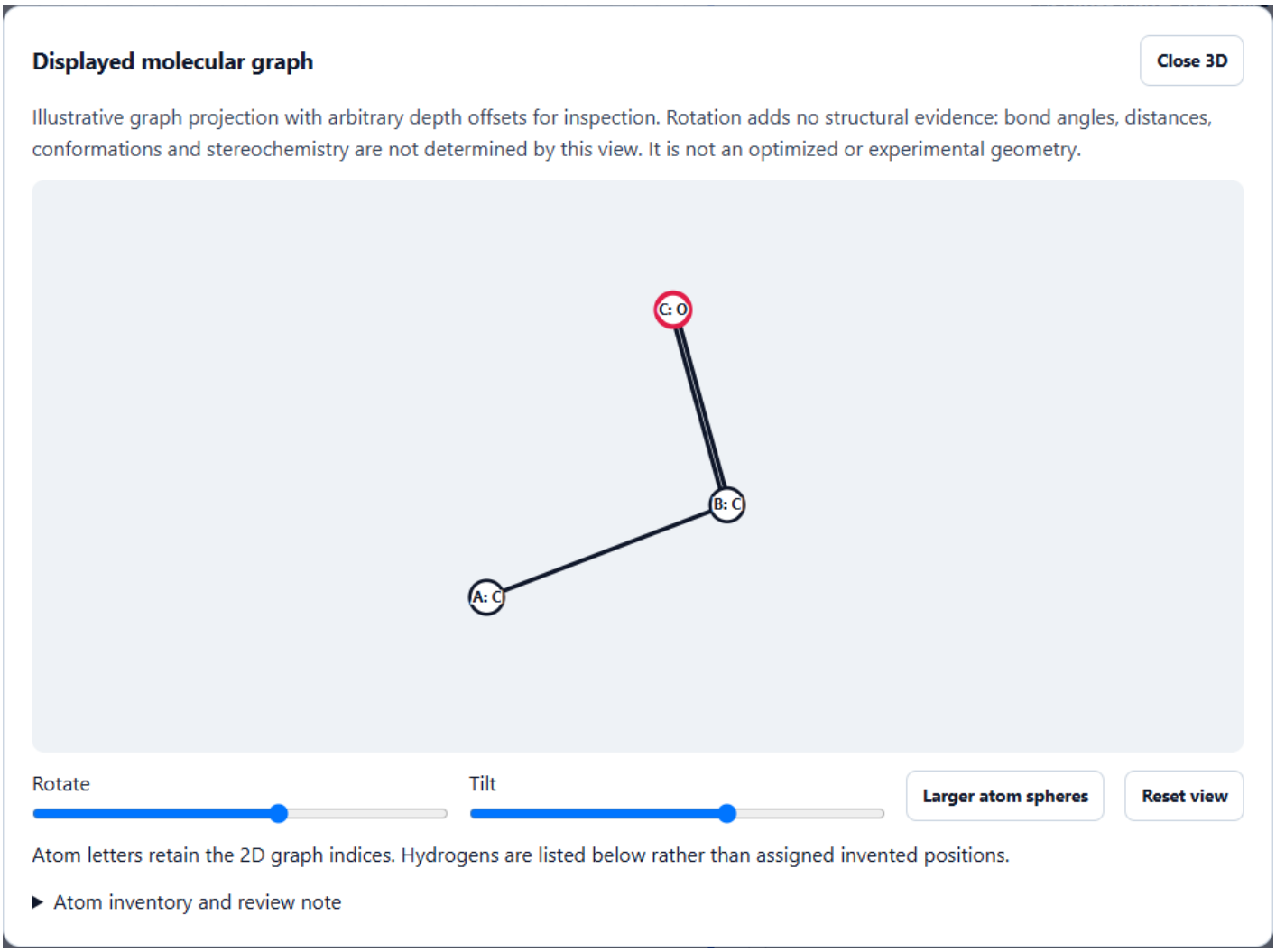
Screenshot: layout-mobile-dark



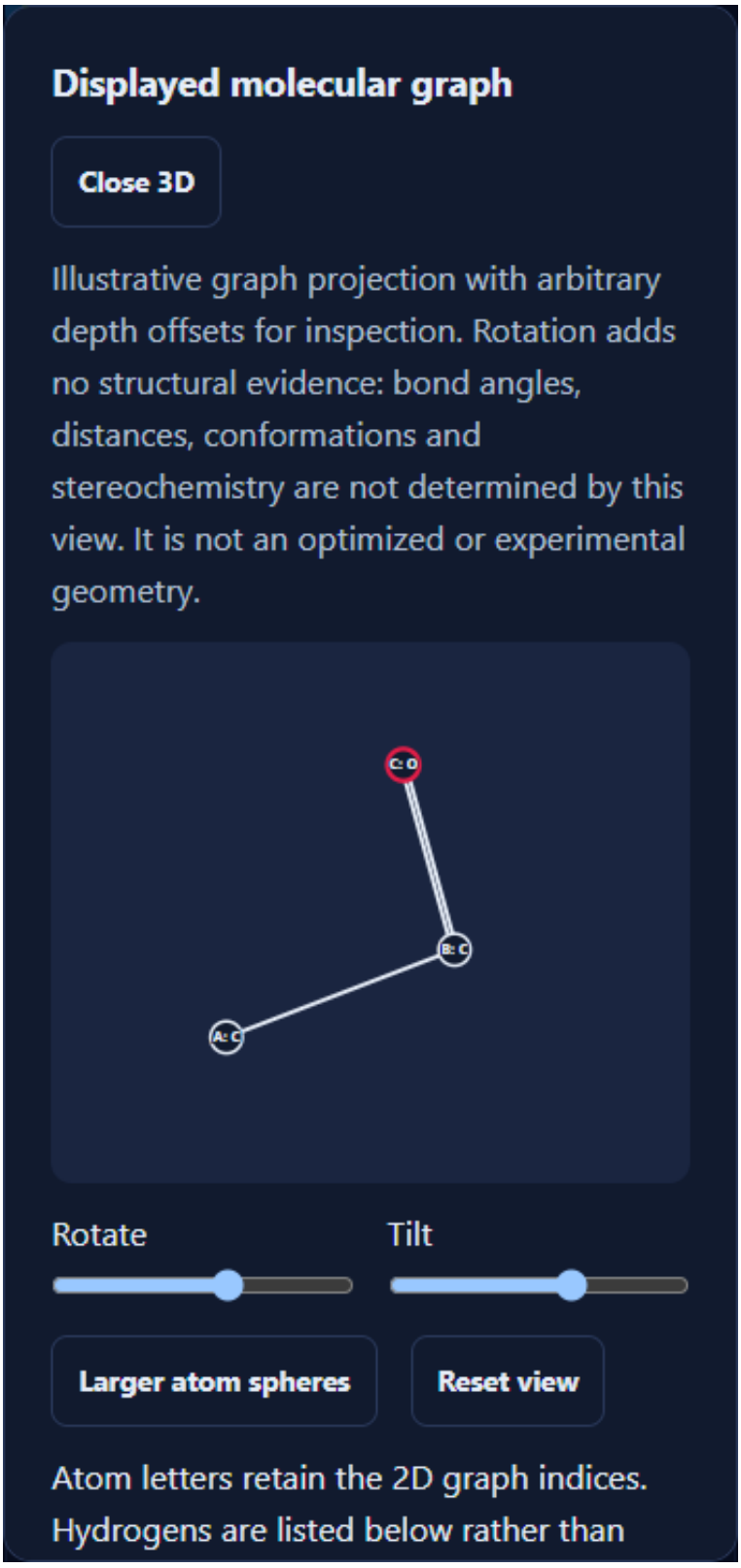
Screenshot: visual-notes-expanded



Screenshot: projection-desktop-light



Screenshot: projection-mobile-dark



Execution evidence

62 screenshots and 13 recorded interactions. Incorrect answers rejected and encoded keys accepted for captured guided stages. This checks implementation against its data; it does not validate chemistry independently. Static screenshots do not verify animation timing, arbitrary clicks, or screen-reader experience.